An Overview of de Rham Cohomology Theory

RS

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1 Introduction: closed forms, exact forms, and a question of Poincaré

1.1 The starting question

A differential form ω on a manifold M is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η. Since $d \circ d = 0$, every exact form is closed, and one can ask the converse: is every closed form exact? The question was raised by Poincaré in 1895 in a form one can state in a first calculus course: a vector field on a domain in $\mathbb{R}^3$ with vanishing curl need not be a gradient; how badly can this fail, and what does the failure measure?

The answer has two halves, one local and one global.

Locally, closed forms are always exact. On a star-shaped open set in $\mathbb{R}^n$ every closed k-form ($k \geq 1$) is d of something; this is the Poincaré lemma, proved in Section 4.2 by writing down the primitive explicitly. The local answer is thus "yes, always".

Globally, the answer is no, and the first counterexample is already the nicest possible one. Let θ be the angular coordinate on $M = \mathbb{R}^2 \setminus \{0\}$. The function θ is defined only locally, up to multiples of 2π, but its differential

$$d\theta = \frac{x\,dy - y\,dx}{x^2 + y^2} \tag{1.1}$$

is a perfectly well-defined 1-form on all of M, and a direct computation gives $d(d\theta) = 0$. Is $d\theta$ exact? If $d\theta = df$ with $f \in C^\infty(M)$, then along the unit circle $S^1 \subset M$, parametrized by $\gamma(t) = (\cos t, \sin t)$,

$$2\pi = \int_0^{2\pi} \gamma^* d\theta = \int_0^{2\pi} \gamma^* df = \int_0^{2\pi} \frac{d}{dt}\big(f(\gamma(t))\big)\,dt = f(1,0) - f(1,0) = 0, \tag{1.2}$$

a contradiction. (The middle step is the chain rule; the last is the fundamental theorem of calculus.) The punchline is that the obstruction is *topological*: it comes from the hole in M, and Stokes' theorem is what ties the analysis to the topology.

This single computation contains the whole subject in miniature. It suggests defining, for each k,

$$H^k(M) \;=\; \frac{\{\text{closed } k\text{-forms on } M\}}{\{\text{exact } k\text{-forms on } M\}}, \tag{1.3}$$

the *de Rham cohomology* of M. Then the computation above says precisely that $H^1(\mathbb{R}^2 \setminus \{0\}) \neq 0$; indeed in Section 4.4 we will see that this group is $\mathbb{R}$, generated by the class of $d\theta/2\pi$.

1.2 What is true, and why it is remarkable

The groups $H^k(M)$ are constructed entirely out of analysis—smooth functions, their derivatives, and their integrals. Nevertheless they depend only on the *homotopy type* of M (Section 4.1), so that $H^k(\mathbb{R}^n) = 0$ for $k \geq 1$ while $H^1(\mathbb{R}^2 \setminus \{0\}) \cong H^1(S^1) \cong \mathbb{R}$. This is already surprising: a purely local calculus object is insensitive to the smooth structure and sees only the coarse shape of the space.

De Rham's theorem, proved in 1931 [8] and made completely general by Weil in 1952 [7], goes much further: the groups defined by (1.3) are canonically isomorphic to the singular cohomology groups $H^k_{\text{sing}}(M; \mathbb{R})$ of algebraic topology. The isomorphism is the obvious pairing, integration:

$$H^k(M) \ni [\omega] \;\longmapsto\; \Big([\sigma] \mapsto \int_\sigma \omega\Big) \in \operatorname{Hom}\big(H_k(M;\mathbb{R}), \mathbb{R}\big). \tag{1.4}$$

We prove this theorem in Section 5 for manifolds admitting finite good covers, which includes every compact manifold, following the double-complex argument of Bott–Tu; the general statement is then quoted with references.

1.3 Why one should care

Three reasons, in increasing order of sophistication.

First, *computability.* Homology theories built on chains require one to understand all possible triangulations or all singular simplices. The de Rham machinery instead answers global questions with local calculus: the cohomology of spheres is computed in a few lines from the Mayer–Vietoris sequence (Sections 4.3 and 7), and the ring structure of $H^*(\mathbb{C}\mathrm{P}^n)$ falls out of Poincaré duality (Section 7.3).

Second, *the product structure.* Forms multiply, so $H^*(M)$ is not just a sequence of vector spaces but a graded-commutative $\mathbb{R}$-algebra, and the product is compatible with everything: pullbacks, the Mayer–Vietoris sequences, Poincaré duality. For $H^*(\mathbb{T}^n)$ the answer is the exterior algebra on n generators; for $H^*(\mathbb{C}\mathrm{P}^n)$ it is a truncated polynomial algebra. This extra structure is the reason de Rham cohomology remains the most practical theory in global analysis.

Third, *the bridge to geometry.* Because the theory is built from derivatives, it interacts with everything else made of derivatives: Riemannian metrics give Hodge theory (each class has a canonical harmonic representative), connections give Chern–Weil theory (curvature forms represent characteristic classes), and the de Rham complex itself is the first nontrivial example of a resolution of the constant sheaf, the prototype of every cohomology theory in complex and algebraic geometry. We return to these pointers in Chapter 8.

1.4 Conventions and notation

Throughout, *manifold* means a smooth manifold, Hausdorff and second-countable, possibly with boundary, and without further mention all maps are smooth and all forms are smooth. Manifolds are assumed connected when a connectedness statement is made; n denotes dimension. We write $\Omega^k(M)$ for the space of smooth k-forms on M, $Z^k(M)$ for the closed ones and $B^k(M)$ for the exact ones, and $H^k(M) = Z^k(M)/B^k(M)$. Cohomology is taken with real coefficients throughout. Compactly supported variants carry a subscript c; they appear in Chapter 6. Our main references are Bott–Tu [1] for the cohomological arguments and Lee [2] for the manifold theory; specific attributions are given inline.

1.5 Organization

Chapter 2 collects the prerequisites and proves Stokes' theorem. Chapter 3 constructs the exterior derivative and the de Rham complex. Chapter 4 contains the three engines of the subject: homotopy invariance, the Poincaré lemma, and Mayer–Vietoris, and uses them to compute the first examples. Chapter 5 proves de Rham's theorem. Chapter 6 develops compactly supported cohomology and Poincaré duality, together with the Künneth formula. Chapter 7 harvests: spheres, tori, complex projective spaces, and the classical applications. Chapter 8 sketches the further directions and closes.

2 Preliminaries: manifolds, forms, integration, Stokes

This chapter collects the background used in the rest of the notes. The experienced reader may skim it, but the sign conventions fixed here—in particular the boundary orientation in §2.6—are used unchanged from Chapter 3 onwards.

2.1 Smooth manifolds

Definition 2.1. A *chart* on a topological space M is a homeomorphism $\varphi : U \to \varphi(U)$ from an open set $U \subset M$ onto an open set $\varphi(U) \subset \mathbb{R}^n$. Two charts $\varphi : U \to \mathbb{R}^n$ and $\psi : V \to \mathbb{R}^n$ are *compatible* if the transition map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is a diffeomorphism of open subsets of $\mathbb{R}^n$ (with C^∞ understood on the closure, where boundary is involved). An *atlas* is a collection of pairwise compatible charts whose domains cover M, and a *smooth structure* is a maximal atlas. A *smooth manifold* is a Hausdorff, second-countable topological space equipped with a smooth structure; if all charts take values in the closed upper half-space $\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}$, one speaks of a manifold *with boundary*, whose boundary ∂M consists of the points mapped to $\partial\mathbb{H}^n = \{x^n = 0\}$ by some (hence any) chart.

A map $f : M \to N$ between manifolds is smooth if $\psi \circ f \circ \varphi^{-1}$ is smooth for all charts; a diffeomorphism is a smooth map with a smooth inverse. The definition of ∂M is sound because a diffeomorphism of open subsets of $\mathbb{H}^n$ preserves $\partial\mathbb{H}^n$; this follows from the inverse function theorem (see Lee [2, Prop. 1.46]).

Example 2.2. (a) $\mathbb{R}^n$ itself, and every open subset of a manifold, with the restricted atlas. (b) The sphere $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$ with stereographic charts: for $N = (0, \ldots, 0, 1)$ and $S = -N$ the maps $x \mapsto (x_1, \ldots, x_n)/(1 \mp x_{n+1})$ define charts $S^n \setminus \{N\} \to \mathbb{R}^n$ and $S^n \setminus \{S\} \to \mathbb{R}^n$ whose transition map is $y \mapsto y/|y|^2$ on $\mathbb{R}^n \setminus \{0\}$, a diffeomorphism. (c) The torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$, with charts obtained by composing local sections of the quotient map $\mathbb{R}^n \to \mathbb{T}^n$. In the language of Exercise 2.3, this is the quotient by a free, properly discontinuous action. (d) The projective spaces $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$: quotients of S^n (respectively $S^{2n+1} \subset \mathbb{C}^{n+1}$) by the antipodal map (respectively by S^1); the standard affine charts $[x_0 : \cdots : x_n] \mapsto (x_0/x_i, \ldots, \widehat{x_i/x_i}, \ldots, x_n/x_i)$ form an atlas of $n + 1$ charts. (e) Products of manifolds, with product charts; graphs of smooth maps $M \to N$; regular level sets (submanifolds), by the implicit function theorem.

Exercise 2.3. Let Γ be a group acting freely and properly discontinuously on a manifold $\widetilde{M}$ by diffeomorphisms. Show that $\widetilde{M}/\Gamma$ carries a unique smooth structure making the quotient map a local diffeomorphism. Apply this to (c) and (d) above.

2.2 Tangent vectors and vector fields

There are several equivalent definitions of the tangent space; the one best suited to forms is the derivations approach.

Definition 2.4. Let $p \in M$ and let $C^\infty(M)$ be the algebra of smooth real-valued functions. A *tangent vector* at p is a linear map $v : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule $v(fg) = v(f)g(p) + f(p)v(g)$. The space of tangent vectors is T_pM, the *tangent space* at p.

In a chart $(x^1, \dots, x^n)$ around p, the vectors $\partial_i|_p = \partial/\partial x^i|_p$ defined by $\partial_i|_p(f) = \partial(f \circ \varphi^{-1})/\partial u^i\big|_{\varphi(p)}$ form a basis of T_pM. This is standard: any derivation annihilates constants, and Taylor's theorem gives $f = f(p) + \sum_i (x^i - x^i(p)) g_i$ with $g_i(p) = \partial_i f(p)$, whence $v(f) = \sum_i v(x^i)\, \partial_i f(p)$. We write $v = \sum_i v^i \partial_i|_p$, $v^i = v(x^i)$.

If $f : M \to N$ is smooth, its *pushforward* (or differential) at p is the linear map $f_{*,p} : T_pM \to T_{f(p)}N$, $(f_{*,p}v)(h) = v(h \circ f)$. The chain rule is the identity $(g \circ f)_* = g_* \circ f_*$, which is immediate from the definition.

A *vector field* on M is a smooth section X of TM: in coordinates $X = \sum_i X^i \partial_i$ with $X^i \in C^\infty$. Vector fields act on functions by $(Xf)(p) = X_p(f)$, producing the directional derivative. The *Lie bracket* of two vector fields is defined by $[X, Y]f = X(Yf) - Y(Xf)$; one checks directly that $[X, Y]$ is again a derivation in f, hence a vector field. In coordinates,

$$[X, Y] = \sum_{i,j} \Big(X^j \frac{\partial Y^i}{\partial x^j} - Y^j \frac{\partial X^i}{\partial x^j} \Big) \partial_i, \tag{2.1}$$

because the second derivatives of f cancel: $X(Yf) - Y(Xf) = \sum_{i,j} (X^j \partial_j Y^i - Y^j \partial_j X^i) \partial_i f + \sum_{i,j} X^j Y^i (\partial_j \partial_i f - \partial_i \partial_j f)$, and the last sum vanishes since mixed partials commute. The bracket is bilinear, antisymmetric, and satisfies the Jacobi identity; these follow from the same identities for commutators of operators.

2.3 The exterior algebra

Let V be a real vector space of dimension n and V^* its dual. A *k-form* on V (more precisely, an alternating k-linear form) is a multilinear map $\alpha : V^k \to \mathbb{R}$ that changes sign under interchange of any two arguments. The space of k-forms is $\Lambda^k V^*$, with $\Lambda^0 V^* = \mathbb{R}$ and $\Lambda^1 V^* = V^*$; for $k > n$ only the zero form exists. If $e_1, \dots, e_n$ is a basis of V with dual basis $e^1, \dots, e^n$, then the forms

$$e^I = e^{i_1} \wedge \cdots \wedge e^{i_k}, \qquad I = (i_1 < i_2 < \cdots < i_k), \tag{2.2}$$

constitute a basis of $\Lambda^k V^*$, so $\dim \Lambda^k V^* = \binom{n}{k}$.

The *wedge product* of $\alpha \in \Lambda^k V^*$ and $\beta \in \Lambda^\ell V^*$ is

$$(\alpha \wedge \beta)(v_1, \dots, v_{k+\ell}) = \frac{1}{k!\,\ell!} \sum_{\sigma \in S_{k+\ell}} \operatorname{sgn}(\sigma)\, \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)})\, \beta(v_{\sigma(k+1)}, \dots, v_{\sigma(k+\ell)}). \tag{2.3}$$

It is bilinear, associative, and *graded commutative*:

$$\beta \wedge \alpha = (-1)^{k\ell}\, \alpha \wedge \beta. \tag{2.4}$$

Associativity is a straightforward but tedious verification from the definition; graded commutativity is seen on basis elements, where $e^{i_1} \wedge \cdots \wedge e^{i_k} \wedge e^{j_1} \wedge \cdots \wedge e^{j_\ell}$ is brought to the order j's first by moving each j-factor across all k of the i-factors, each crossing contributing one sign change, in total $(-1)^{k\ell}$. We will use (2.4) constantly and without comment.

The *interior product* with a vector $v \in V$ is the operator $\iota_v : \Lambda^k V^* \to \Lambda^{k-1} V^*$ defined by

$$(\iota_v \alpha)(w_1, \dots, w_{k-1}) = \alpha(v, w_1, \dots, w_{k-1}). \tag{2.5}$$

The identity

$$\iota_v(\alpha \wedge \beta) = (\iota_v \alpha) \wedge \beta + (-1)^k \alpha \wedge (\iota_v \beta) \tag{2.6}$$

holds for $\alpha \in \Lambda^k V^*$, $\beta \in \Lambda^\ell V^*$. It is checked on basis elements: the terms in which v is fed to α (respectively to β) account for the two summands, and the sign $(-1)^k$ appears because v must move past the k factors of α before reaching β. Both sides are bilinear in (α, β), so basis elements suffice.

2.4 Differential forms and pullback

Definition 2.5. Let M be a manifold. A *differential k-form* on M is a smooth assignment $p \mapsto \omega_p \in \Lambda^k(T_pM)^*$; smoothness means that in any chart the coefficients of ω in the basis $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ are smooth functions. The real vector space of k-forms is denoted $\Omega^k(M)$, with $\Omega^0(M) = C^\infty(M)$; the direct sum is $\Omega^*(M) = \bigoplus_k \Omega^k(M)$.

In a chart, every k-form is uniquely

$$\omega = \sum_{i_1 < \cdots < i_k} \omega_{i_1 \dots i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \sum_I \omega_I \, dx^I, \tag{2.7}$$

with $\omega_I \in C^\infty$. The wedge product extends pointwise to $\Omega^*(M)$ and makes it a graded-commutative algebra over $\mathbb{R}$, the identity (2.4) holding fibre by fibre.

The *pullback* of a form under a smooth map $f : M \to N$ is defined pointwise by

$$(f^*\omega)_p(v_1, \dots, v_k) = \omega_{f(p)}(f_{*,p}v_1, \dots, f_{*,p}v_k), \tag{2.8}$$

so that $f^* : \Omega^k(N) \to \Omega^k(M)$ is linear, and for functions $f^*h = h \circ f$. From the definitions one obtains, exactly as for the interior product,

$$f^*(\alpha \wedge \beta) = (f^*\alpha) \wedge (f^*\beta), \qquad (g \circ f)^* = f^* \circ g^*. \tag{2.9}$$

Both identities are verified by evaluating on tangent vectors: the first is the pointwise algebra identity, and the second follows from the chain rule $(g \circ f)_* = g_* \circ f_*$. Note the reversal: the pullback is contravariant, which is what will make cohomology a contravariant functor.

2.5 Partitions of unity

Partitions of unity are the basic gluing device of manifold theory. They rest on the following elementary construction.

Lemma 2.6 (Bump functions)**.** *For every $0 < r < R$ there exists a smooth function $\rho : \mathbb{R}^n \to [0, 1]$ with $\rho \equiv 1$ on the closed ball $\overline{B}_r(0)$ and $\operatorname{supp} \rho \subset B_R(0)$.*

Proof. The classical one-dimensional function is $h(t) = e^{-1/t}$ for $t > 0$ and $h(t) = 0$ for $t \leq 0$, which is smooth: the only question is at $t = 0$, where every derivative of h is a finite linear combination of terms $t^{-m}e^{-1/t}$, each tending to 0 as $t \to 0^+$, so all derivatives of h exist and vanish at 0; smoothness follows by induction. On $\mathbb{R}^n$ set

$$g(x) = \frac{h\big(R^2 - |x|^2\big)}{h\big(R^2 - |x|^2\big) + h\big(|x|^2 - r^2\big)}$$

is smooth on $\mathbb{R}^n$: the denominator never vanishes, since $|x|^2 - r^2 > 0$ and $R^2 - |x|^2 > 0$ cannot hold simultaneously for $r < R$ (one of the two arguments of h is positive, the other nonpositive). For $|x| \leq r$ the first numerator term is constant $h(R^2 - r^2) > 0$ while $h(|x|^2 - r^2) = 0$, so $g = 1$; for $|x| \geq R$ we get $g = 0$; and $0 \leq g \leq 1$ throughout. This g is ρ. $\square$

Theorem 2.7 (Partitions of unity)**.** *Let $\mathcal{U} = \{U_\alpha\}$ be an open cover of a manifold M. There exist smooth functions $\rho_\alpha : M \to [0, 1]$ such that*

(i) $\operatorname{supp} \rho_\alpha \subset U_\alpha$ *for all α;*

(ii) the family $\{\operatorname{supp} \rho_\alpha\}$ *is locally finite, i.e. every point of M has a neighbourhood meeting only finitely many of them;*

(iii) $\sum_\alpha \rho_\alpha \equiv 1$ *on M.*

If M is compact, finitely many ρ_α suffice.

Proof. We give the proof for the case needed later; the general statement differs only in bookkeeping. Since M is second-countable and locally compact, it is covered by a countable, locally finite family of precompact chart domains W_j, $\overline{W}_j$ compact, with each $\overline{W}_j$ contained in some $U_{\alpha(j)}$. (Cover M by countably many charts; take the precompact refinement; local finiteness is automatic from second-countability, see Lee [2, Thm. 2.19].) For each j choose, using Lemma 2.6 in the chart for W_j, a smooth $\chi_j \geq 0$ with $\chi_j > 0$ on W_j and $\operatorname{supp} \chi_j \subset U_{\alpha(j)}$. The sum $\chi = \sum_j \chi_j$ is smooth because the family is locally finite, and $\chi > 0$ everywhere. Put

$$\rho_\alpha = \frac{1}{\chi} \sum_{j:\alpha(j)=\alpha} \chi_j.$$

Then $\operatorname{supp} \rho_\alpha \subset U_\alpha$ (finite unions of closed sets, again locally finite), and $\sum_\alpha \rho_\alpha = \chi/\chi = 1$. □

Remark 2.8. The same construction yields partitions of unity subordinate to any locally finite refinement of $\mathcal{U}$; in particular, given two open sets U, V with $M = U \cup V$, we will repeatedly use functions $\rho_U, \rho_V \in C^\infty(M)$ with $\operatorname{supp} \rho_U \subset U$, $\operatorname{supp} \rho_V \subset V$ and $\rho_U + \rho_V = 1$.

2.6 Orientation and integration

A *pointwise orientation* of a vector space V is an equivalence class of bases under the relation "same orientation", i.e. a choice of sign of the determinant of the change-of-basis matrix. A manifold M is *orientable* if there is a continuous pointwise orientation of each T_pM; an *orientation* is such a choice. Equivalently (and this is the form we shall use), M is orientable if and only if it admits a nowhere-vanishing n-form $\mu \in \Omega^n(M)$, called a *volume form*: a pointwise orientation is then declared by requiring $\mu_p(v_1, \ldots, v_n) > 0$ on oriented bases. The equivalence of the two definitions is standard; one direction is immediate (a volume form orients each fibre), the other uses a partition of unity to patch local volume forms with coherent signs (see Lee [2, Prop. 15.5]). An *orientation-preserving* diffeomorphism $f : M \to N$ is one for which f_* carries the orientation of each T_pM to that of $T_{f(p)}N$; equivalently $f^*\nu = e^h\mu$ with $h \in C^\infty(M)$ for any volume forms μ on M, ν on N.

Integration is defined in three steps.

Step 1: forms supported in one chart. Let $\omega \in \Omega^n(M)$ with $\operatorname{supp} \omega$ contained in the domain of an oriented chart $\varphi : U \to \widehat{U} \subset \mathbb{R}^n$, and write $(\varphi^{-1})^*\omega = f\, dx^1 \wedge \cdots \wedge dx^n$ with $f \in C_c^\infty(\widehat{U})$. Set

$$\int_M \omega = \int_{\widehat{U}} f(x^1, \ldots, x^n)\, dx^1 \ldots dx^n, \tag{2.10}$$

the ordinary (improper) Riemann or Lebesgue integral.

Step 2: the general case. Let $\{(U_\alpha, \varphi_\alpha)\}$ be an oriented atlas and $\{\rho_\alpha\}$ a partition of unity subordinate to it (Theorem 2.7). Define

$$\int_M \omega = \sum_\alpha \int_{U_\alpha} \rho_\alpha \omega, \tag{2.11}$$

each summand computed as in Step 1; the sum is finite whenever $\operatorname{supp} \omega$ is compact, which is the only case we integrate. For the definition to make sense one must know that the right-hand side is independent of the atlas and the partition of unity. This is a consequence of the change-of-variables theorem: if φ, ψ are two oriented charts with overlap V, then on $\varphi(V \cap U \cap U_\psi)$ the two local representatives of a form are related by the (orientation-preserving, hence positive-Jacobian) transition map, and the classical formula

$$\int_{\psi(V)} f(y)\, dy = \int_{\varphi(V)} f\big(\psi \circ \varphi^{-1}(x)\big) \left|\det D(\psi \circ \varphi^{-1})\right| dx$$

shows the two computations agree. We shall use the change-of-variables theorem as a black box from calculus; its use here and in Proposition 2.9 is the only place where it enters.

Step 3: boundary orientation. If $(M, \partial M)$ is an oriented manifold with boundary, ∂M receives an induced orientation as follows. At $p \in \partial M$ choose a boundary chart $(x^1, \dots, x^{n-1}, x^n)$ with $x^n \geq 0$ on M; the basis $(\partial_1, \dots, \partial_{n-1}, \partial_n)$ of T_pM is oriented if and only if ∂_n points *outward*. Then the basis $(\partial_1, \dots, \partial_{n-1})$ of $T_p\partial M$ is declared oriented when $(\partial_n, \partial_1, \dots, \partial_{n-1})$ is positively oriented in T_pM. Equivalently: in terms of forms, if $dx^1 \wedge \cdots \wedge dx^n$ is the positive volume form, the induced volume form on the boundary is

$$\omega_{\partial M} = (-1)^n \, dx^1 \wedge \cdots \wedge dx^{n-1} \tag{2.12}$$

(restricted to ∂M, where $x^n = 0$), since $\iota_{\partial_n}(dx^1 \wedge \cdots \wedge dx^n) = (-1)^{n-1} dx^1 \wedge \cdots \wedge dx^{n-1}$ and the sign $(-1)^n$ records that ∂_n was moved from first to last position. The sign convention (2.12) is the one of Bott–Tu and Lee; it makes the boundary of the unit interval $\partial[0,1]$ positively oriented at $t = 1$ and negatively at $t = 0$, which is what makes the fundamental theorem of calculus come out as $\int_{[0,1]} df = f(1) - f(0) = \int_{\partial[0,1]} f$.

Proposition 2.9. *Let $f : M \to N$ be an orientation-preserving diffeomorphism of oriented n-manifolds. Then for every $\omega \in \Omega^n_c(N)$, $\int_M f^*\omega = \int_N \omega$.*

Proof. By a partition of unity on N it suffices to consider ω supported in one chart of N; then $f^*\omega$ is supported in the corresponding chart of M (taking the chart pulled back along f), and the claim is exactly the change-of-variables formula for the orientation-preserving local representative of f. □

2.7 Stokes' theorem

Theorem 2.10 (Stokes)**.** *Let M be an oriented smooth n-manifold with boundary ∂M (with the induced orientation (2.12)). For every $\omega \in \Omega^{n-1}_c(M)$,*

$$\int_M d\omega = \int_{\partial M} \omega, \tag{2.13}$$

where the right-hand side means $\int_{\partial M} \omega|_{\partial M}$.

Proof. The proof is a reduction to the fundamental theorem of calculus, which is where all the content lives.

Reduction 1: partition of unity. Choose finitely many charts $\{(U_\alpha, \varphi_\alpha)\}$ covering $\operatorname{supp}\omega$ and a subordinate partition of unity ρ_α. By linearity of both sides of (2.13) it is enough to prove the theorem for each $\rho_\alpha\omega$, i.e. we may assume $\operatorname{supp}\omega$ is contained in one chart domain U.

Reduction 2: the local computation. Pull the problem back to $\mathbb{R}^n$ or $\mathbb{H}^n$: it suffices to prove the following two statements.

Case A (interior chart). If $\omega = \sum_i f_i \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$ with $f_i \in C^\infty_c(\mathbb{R}^n)$, then $\int_{\mathbb{R}^n} d\omega = 0$. Indeed $d\omega = \sum_i (-1)^{i-1} (\partial f_i / \partial x^i) \, dx^1 \wedge \cdots \wedge dx^n$, and

$$\int_{\mathbb{R}^n} \frac{\partial f_i}{\partial x^i}(x) \, dx = \int_{\mathbb{R}^{n-1}} \Big(\int_{-\infty}^{\infty} \frac{\partial f_i}{\partial x^i}(x) \, dx^i \Big) dx^1 \dots \widehat{dx^i} \dots dx^n = 0$$

by Fubini and the fundamental theorem of calculus, since f_i has compact support in the x^i direction.

Case B (boundary chart). Let $\omega = \sum_{i=1}^n f_i \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$ with $f_i \in C^\infty_c(\mathbb{H}^n)$. For $i < n$ the same computation as in Case A gives zero. For $i = n$, writing $x' = (x^1, \dots, x^{n-1})$:

$$\begin{aligned} \int_{\mathbb{H}^n} d\big(f_n \, dx^1 \wedge \cdots \wedge dx^{n-1}\big) &= (-1)^{n-1} \int_{\mathbb{R}^{n-1}} \Big(\int_0^\infty \frac{\partial f_n}{\partial x^n} \, dx^n \Big) dx' \\ &= (-1)^{n-1} \int_{\mathbb{R}^{n-1}} \big(0 - f_n(x', 0)\big) \, dx' = (-1)^n \int_{\mathbb{R}^{n-1}} f_n(x', 0) \, dx'. \end{aligned} \tag{2.14}$$

On the other hand, the restriction of ω to $\partial\mathbb{H}^n = \{x^n = 0\}$ is $f_n(x', 0)\, dx^1 \wedge \cdots \wedge dx^{n-1}$, and by the boundary orientation convention (2.12) the induced oriented volume form on $\partial\mathbb{H}^n$ is exactly $(-1)^n dx^1 \wedge \cdots \wedge dx^{n-1}$. Hence $\int_{\mathbb{H}^n} d\omega = \int_{\partial\mathbb{H}^n} \omega$, which is (2.13) in the model case.

Conclusion. By Reduction 1 and Reduction 2, $\int_M d\omega = \sum_\alpha \int \varphi_\alpha^* d(\rho_\alpha \omega) = \sum_\alpha \int d(\varphi_\alpha^* \rho_\alpha \omega)$, and each summand equals the boundary integral in the chart; the sum of the boundary integrals is $\int_{\partial M} \omega$ by the definition of integration on ∂M. (The pullback of the orientation form brings along exactly the signs of Case B; this is where the convention (2.12) earns its keep.) □

Remark 2.11. Stokes' theorem holds verbatim for forms whose support meets ∂M compactly; we only ever need the compact case. The theorem is also the single place where the smoothness of everything is used in an essential way: all later chapters are, in a sense, elaborate bookkeeping around (2.13).

Exercise 2.12. Recover the three classical theorems of vector calculus on $\mathbb{R}^3$ (Gauss, classical Stokes, Green) by writing vector fields as 1- and 2-forms via the metric and the volume form, and applying Theorem 2.10 to a ball, a surface with boundary, and a planar domain.

3 The exterior derivative and the de Rham complex

3.1 Motivation

The differential of a function $f \in C^\infty(M)$ is the 1-form df defined by $df(v) = v(f)$; in coordinates $df = \sum_j (\partial_j f) dx^j$. We want an analogue for forms of all degrees: an operation $d : \Omega^k(M) \to \Omega^{k+1}(M)$ that generalizes df, satisfies $d \circ d = 0$, and obeys a Leibniz rule. The pleasant fact, proved below, is that these three requirements determine d completely; there is no choice of "which derivative to use".

3.2 Existence, uniqueness, and the invariant formula

In a coordinate chart, define

$$d\Big(\sum_I f_I\, dx^I\Big) = \sum_I df_I \wedge dx^I = \sum_I \sum_j \frac{\partial f_I}{\partial x^j}\, dx^j \wedge dx^I. \tag{3.1}$$

This is a perfectly good operator in each chart, but it is not clear that charts agree on overlaps. The next theorem resolves this and, in passing, exhibits the coordinate-free meaning of d.

Theorem 3.1 (Invariant formula for d)**.** *There is a unique family of linear operators $d : \Omega^k(M) \to \Omega^{k+1}(M)$, $k \geq 0$, such that*

(i) df is the differential of f for $f \in \Omega^0(M)$;

(ii) $d \circ d = 0$;

(iii) $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ for $\alpha \in \Omega^k(M)$.

It is given on vector fields $X_0, \ldots, X_k$ by

$$\begin{aligned} d\omega(X_0, \ldots, X_k) = &\sum_{i=0}^{k} (-1)^i X_i\big(\omega(X_0, \ldots, \widehat{X_i}, \ldots, X_k)\big) \\ &+ \sum_{0 \leq i < j \leq k} (-1)^{i+j}\, \omega\big([X_i, X_j], X_0, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_k\big). \end{aligned} \tag{3.2}$$

Proof. Uniqueness. Suppose d satisfies (i)–(iii). For a function g and a 1-form α, property (iii) gives $d(g\alpha) = dg \wedge \alpha + g\, d\alpha$; iterating over the factors of a wedge, and applying (ii) to each $d(dx^i) = d^2 x^i = 0$, yields for every multi-index $I = (i_1 < \cdots < i_k)$:

$$d(dx^I) = d\big(dx^{i_1} \wedge \cdots \wedge dx^{i_k}\big) = \sum_{\nu=1}^{k} \pm\, dx^{i_1} \wedge \cdots \wedge d(dx^{i_\nu}) \wedge \cdots \wedge dx^{i_k} = 0,$$

the signs being irrelevant since every summand vanishes. Consequently

$$d\Big(\sum_I f_I\, dx^I\Big) = \sum_I d(f_I\, dx^I) = \sum_I \big(df_I \wedge dx^I + f_I\, d(dx^I)\big) = \sum_I df_I \wedge dx^I,$$

so (i)–(iii) force the local expression (3.1). This proves uniqueness.

Existence. Define d chart by chart via (3.1). Properties (i) and (iii) are immediate from the definition, and (ii) is the computation

$$d^2(f\, dx^I) = d\Big(\sum_j \frac{\partial f}{\partial x^j}\, dx^j \wedge dx^I\Big) = \sum_{i,j} \frac{\partial^2 f}{\partial x^i \partial x^j}\, dx^i \wedge dx^j \wedge dx^I = 0,$$

the sum vanishing termwise because $\partial_i \partial_j f = \partial_j \partial_i f$ while $dx^i \wedge dx^j = -dx^j \wedge dx^i$. Two charts agree on their overlap, because on $U \cap V$ both operators satisfy (i)–(iii), and uniqueness (already proved) applies to the manifold $U \cap V$.

The invariant formula. It remains to identify d with the right-hand side of (3.2). Both sides are $\mathbb{R}$-linear in ω, so we may work in one chart and take $\omega = f\,\eta$ with $\eta = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ and $f \in C^\infty$. The following cancellation is the heart of the matter.

Lemma 3.2. *For $\eta = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ and vector fields $X_0, \ldots, X_k$ on the chart domain,*

$$\begin{aligned} &\sum_{i=0}^{k} (-1)^i\, X_i\big(\eta(X_0, \ldots, \widehat{X_i}, \ldots, X_k)\big) \\ &= -\sum_{0 \le i < j \le k} (-1)^{i+j}\, \eta\big([X_i, X_j], X_0, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_k\big). \end{aligned} \tag{3.3}$$

Proof of the lemma. Write $a_{\alpha j} = dx^{i_\alpha}(X_j)$; this is a $k \times (k+1)$ matrix of functions, and $\eta(X_0, \ldots, \widehat{X_i}, \ldots, X_k) = \det M^{(i)}$, where $M^{(i)}$ is obtained from $M = (a_{\alpha j})$ by deleting column i. Differentiating a determinant column by column, then moving the differentiated column to the front (it stands at position j among the remaining columns when $j < i$, and at position $j-1$ when $j > i$), gives

$$\sum_i (-1)^i X_i\big(\det M^{(i)}\big) = \sum_{i<j} (-1)^{i+j-1} D_{ij} + \sum_{i>j} (-1)^{i+j} D_{ij}, \tag{3.4}$$

where D_{ij} is the determinant of the matrix whose first column is $(X_i a_{1j}, \ldots, X_i a_{kj})$ and whose remaining columns are the columns of M other than i and j, in increasing order. On the other hand $dx^{i_\alpha}([X_i, X_j]) = [X_i, X_j]^{i_\alpha} = X_i a_{\alpha j} - X_j a_{\alpha i}$ by the bracket formula (2.1), so

$$\sum_{i<j} (-1)^{i+j}\, \eta\big([X_i, X_j], X_0, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_k\big) = \sum_{i<j} (-1)^{i+j} \big(D_{ij} - E_{ij}\big), \tag{3.5}$$

with E_{ij} built like D_{ij} but from the column $(X_j a_{1i}, \ldots, X_j a_{ki})$. Since $E_{ij} = D_{ji}$, the sum $\sum_{i>j} (-1)^{i+j} D_{ij}$ reindexed as a sum over pairs with $j < i$ equals $\sum_{i<j} (-1)^{i+j} E_{ij}$, and adding (3.4) and (3.5) gives zero. $\square$

Returning to $\omega = f\eta$: by (3.1) and the pairing identity for a wedge of a 1-form with a k-form,

$$d\omega(X_0,\dots,X_k) = (df\wedge\eta)(X_0,\dots,X_k) = \sum_i (-1)^i\, df(X_i)\,\eta(X_0,\dots,\widehat{X_i},\dots,X_k).$$

Write $\eta_i = \eta(X_0,\dots,\widehat{X_i},\dots,X_k)$ and $\eta_{ij} = \eta\big([X_i,X_j],X_0,\dots,\widehat{X_i},\dots,\widehat{X_j},\dots,X_k\big)$. The right-hand side of (3.2), evaluated on $\omega = f\eta$, is

$$\sum_i (-1)^i X_i\big(f\,\eta_i\big) + \sum_{i<j}(-1)^{i+j} f\,\eta_{ij} = \sum_i (-1)^i (X_i f)\,\eta_i + f\Big[\sum_i (-1)^i X_i\eta_i + \sum_{i<j}(-1)^{i+j}\eta_{ij}\Big],$$

and the bracket vanishes by Lemma 3.2. Since $df(X_i) = X_i f$, the two sides agree, as claimed. Because the right-hand side of (3.2) thus equals the form $d\omega$, it is in particular C^∞-multilinear and alternating in the vector fields $X_0,\dots,X_k$, and it is independent of the choice of chart, since d is. □

Remark 3.3. The case $k = 1$ of (3.2),

$$d\omega(X,Y) = X\omega(Y) - Y\omega(X) - \omega([X,Y]), \tag{3.6}$$

is used so often that it deserves a name of its own. In vector-calculus language it says: the curl of a vector field is measured by the failure of the field to be a gradient, corrected for the noncommutativity of the coordinate vector fields—on $\mathbb{R}^n$ the bracket term vanishes and one recovers curl.

3.3 Basic properties

Proposition 3.4. *The exterior derivative satisfies:*

(i) $d^2 = 0$*;*

(ii) $d(\alpha\wedge\beta) = d\alpha\wedge\beta + (-1)^{\deg\alpha}\alpha\wedge d\beta$ *(the* antiderivation *property);*

(iii) naturality: for every smooth $f : M \to N$, $d(f^*\omega) = f^*(d\omega)$ *for all* $\omega \in \Omega^k(N)$.

Proof. (i) and (ii) were verified in the proof of Theorem 3.1 (they hold chart by chart by (3.1)). For (iii) it suffices, by linearity, to consider $\omega = g\,dy^I$ in a chart on N (with f^{-1} of the chart domain as chart on M). Then $f^*\omega = (g\circ f)\,d(y^{i_1}\circ f)\wedge\cdots\wedge d(y^{i_k}\circ f)$ by (2.9), so

$$d(f^*\omega) = d(g\circ f)\wedge d(y^{i_1}\circ f)\wedge\cdots\wedge d(y^{i_k}\circ f) = f^*(dg)\wedge f^*(dy^{i_1})\wedge\cdots = f^*(dg\wedge dy^I) = f^*(d\omega),$$

where the first equality uses (ii) and the fact that $d^2(y^i\circ f) = 0$, the second uses the chain rule for functions and the third uses (2.9) again. □

3.4 The de Rham complex

Definition 3.5. The *de Rham complex* of M is the cochain complex

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots, \tag{3.7}$$

i.e. the sequence of vector spaces and linear maps with $d^2 = 0$. We write

$$Z^k(M) = \ker\big(d : \Omega^k(M) \to \Omega^{k+1}(M)\big), \qquad B^k(M) = \operatorname{im}\big(d : \Omega^{k-1}(M) \to \Omega^k(M)\big),$$

for the spaces of *closed* and *exact* k-forms (with $B^0(M) = 0$). The *k-th de Rham cohomology* of M is the quotient

$$H^k(M) = H^k_{\mathrm{dR}}(M) = Z^k(M)/B^k(M). \tag{3.8}$$

The cochain-complex language is not decoration: the functorial behaviour of cohomology, the long exact sequences of Chapter 4, and the double complexes of Chapter 5 all live most naturally at the level of complexes, with H^k applied at the end. A *cochain map* between complexes is a family of linear maps $\varphi^k : A^k \to B^k$ with $d\varphi^k = \varphi^{k+1}d$; it sends closed forms to closed forms and exact forms to exact forms, hence induces maps $H^k(\varphi) : H^k(A) \to H^k(B)$.

By Proposition 3.4(iii), every smooth map $f : M \to N$ induces a cochain map $f^* : \Omega^*(N) \to \Omega^*(M)$, and hence linear maps

$$f^* : H^k(N) \longrightarrow H^k(M), \qquad [\omega] \mapsto [f^*\omega]. \tag{3.9}$$

The identities (2.9) make H^k a contravariant functor from smooth manifolds to real vector spaces: $(g \circ f)^* = f^* \circ g^*$ and $\mathrm{id}^* = \mathrm{id}$. In particular, diffeomorphic manifolds have isomorphic de Rham cohomology. Chapter 4 strengthens this considerably.

3.5 The first computations

Proposition 3.6. *For any manifold M, $H^0(M) \cong \mathbb{R}^{\pi_0(M)}$, the space of locally constant real functions. If M is connected, $H^0(M) = \mathbb{R}$ with generator* $[1]$.

Proof. $Z^0(M)$ consists of functions f with $df = 0$. In a chart, $df = \sum_j \partial_j f\, dx^j$ vanishes if and only if all partial derivatives vanish, i.e. f is locally constant. A locally constant function is constant on each connected component, so $Z^0(M) = \mathbb{R}^{\pi_0(M)}$. Since $B^0(M) = 0$, the claim follows. □

Proposition 3.7. $H^1(\mathbb{R}) = 0$*, and more generally $H^1(I) = 0$ for any interval I.*

Proof. Every 1-form on $\mathbb{R}$ is $\omega = g(x)\, dx$ with $g \in C^\infty(\mathbb{R})$, and every 1-form is automatically closed. Define $f(x) = \int_0^x g(t)\, dt$; then $df = g\, dx = \omega$ by the fundamental theorem of calculus, so ω is exact. □

The same argument computes $H^1(S^1)$, but there the primitive $\int_0^\theta g$ is defined only up to the total integral of g around the circle; this is the prototype of the obstruction mechanism. We postpone the computation to Section 4.4, after the Mayer–Vietoris machine is in place.

Exercise 3.8. (a) Show directly from the definitions that every closed 1-form on $\mathbb{R}^2$ is exact. (b) Let $\omega = (x\, dy - y\, dx)/(x^2 + y^2)$ on $\mathbb{R}^2 \setminus \{0\}$. Verify $d\omega = 0$ and show that $\omega = d\theta$ in polar coordinates; conclude with Stokes' theorem, as in the introduction, that ω is not exact.

4 Homotopy invariance, the Poincaré lemma, and Mayer–Vietoris

This chapter builds the three engines that make de Rham cohomology computable. The first says that the cohomology depends only on the homotopy type; the second says that locally all cohomology is trivial; the third assembles local information into global information. Everything else in the notes is a consequence of these three facts.

4.1 Homotopy invariance

Theorem 4.1 (Homotopy invariance)**.** *Let $f, g : M \to N$ be smooth maps. If f and g are smoothly homotopic, then the induced maps on de Rham cohomology agree: $f^* = g^* : H^k(N) \to H^k(M)$ for all k.*

Proof. Let $F : M \times [0,1] \to N$ be a smooth homotopy from $f = F \circ i_0$ to $g = F \circ i_1$, where $i_t(x) = (x, t)$, and let $\pi : M \times [0,1] \to M$ be the projection. The strategy is the usual one in homological algebra: produce a linear operator $K : \Omega^k(M \times [0,1]) \to \Omega^{k-1}(M)$ satisfying

$$dK + Kd = i_1^* - i_0^*. \tag{4.1}$$

Then $h := K \circ F^* : \Omega^k(N) \to \Omega^{k-1}(M)$ satisfies

$$dh + hd = (dK + Kd) \circ F^* = (i_1^* - i_0^*) \circ F^* = (F \circ i_1)^* - (F \circ i_0)^* = g^* - f^*, \qquad (4.2)$$

using the naturality of d (Proposition 3.4). A linear map h with $dh + hd = g^* - f^*$ is called a *cochain homotopy* from f^* to g^*; if $\omega \in Z^k(N)$, then $g^*\omega - f^*\omega = dh\omega$ is exact, so $g^*[\omega] = f^*[\omega]$.

It remains to construct K. In local coordinates $(x^1, \ldots, x^n, t)$ on $M \times [0, 1]$, every k-form is uniquely

$$\omega = \sum_I f_I(x,t)\, dt \wedge dx^I + \sum_J g_J(x,t)\, dx^J,$$

where the multi-indices in the first sum have length $k-1$ and those in the second have length k. Define

$$(K\omega)(x) = \sum_I \Big(\int_0^1 f_I(x,t)\, dt \Big) dx^I. \qquad (4.3)$$

One must check that this is independent of coordinates, i.e. that (4.3) patches to a global operator; this follows from the coordinate-free description $(K\omega)_x(v_1, \ldots, v_{k-1}) = \int_0^1 \omega_{(x,t)}\big(\partial_t, \widetilde{v}_1, \ldots, \widetilde{v}_{k-1}\big)\, dt$, where $\widetilde{v}_j$ is the horizontal lift of v_j to $T_{(x,t)}(M \times [0, 1])$ and ∂_t is the unit vector in the interval direction.

Now compute, in coordinates:

$$dK\omega = \sum_I \sum_j \Big(\int_0^1 \frac{\partial f_I}{\partial x^j}(x,t)\, dt \Big) dx^j \wedge dx^I, \qquad (4.4)$$

$$d\omega = \sum_J \Big(\frac{\partial g_J}{\partial t}\, dt \wedge dx^J + \sum_j \frac{\partial g_J}{\partial x^j}\, dx^j \wedge dx^J \Big) + \sum_I \sum_j \frac{\partial f_I}{\partial x^j}\, dx^j \wedge dt \wedge dx^I, \qquad (4.5)$$

$$Kd\omega = \sum_J \Big(\int_0^1 \frac{\partial g_J}{\partial t}(x,t)\, dt \Big) dx^J - \sum_I \sum_j \Big(\int_0^1 \frac{\partial f_I}{\partial x^j}(x,t)\, dt \Big) dx^j \wedge dx^I. \qquad (4.6)$$

Here the signs deserve a word. In the second line, the term $\sum_{I,j} \partial_j f_I\, dx^j \wedge dt \wedge dx^I$ comes from $df_I \wedge dt \wedge dx^I$ (the $dt \wedge dt$ piece vanishes), and its contribution to K in the third line carries the sign $-$, because $dx^j \wedge dt = -dt \wedge dx^j$. Adding the first and third displays, the double sums cancel, leaving

$$(dK + Kd)\omega = \sum_J \big(g_J(x,1) - g_J(x,0) \big) dx^J = (i_1^* - i_0^*)\omega,$$

because $i_t^*\omega = \sum_J g_J(x,t)\, dx^J$. This proves (4.1). □

Corollary 4.2. *If M and N are homotopy equivalent, then $H^k(M) \cong H^k(N)$ for all k. In particular, if M is contractible, then $H^0(M) = \mathbb{R}$ and $H^k(M) = 0$ for $k \geq 1$.*

Proof. Let $\varphi : M \to N$ and $\psi : N \to M$ be smooth maps with $\psi \circ \varphi \simeq \mathrm{id}_M$ and $\varphi \circ \psi \simeq \mathrm{id}_N$. By Theorem 4.1,

$$\varphi^* \circ \psi^* = (\psi \circ \varphi)^* = \mathrm{id}_{H^*(M)}, \qquad \psi^* \circ \varphi^* = (\varphi \circ \psi)^* = \mathrm{id}_{H^*(N)},$$

so φ^* and ψ^* are mutually inverse isomorphisms. □

Remark 4.3. The proof contains the whole geometric content of the theorem: the operator K integrates along the homotopy parameter, and (4.1) is the fundamental theorem of calculus with boundary terms at $t = 0, 1$. Everything else is bookkeeping. Note also that *smooth* homotopy suffices: any continuous homotopy between smooth maps can be smoothed relative to the endpoints (see Lee [2, Thm. 6.29]).

4.2 The Poincaré lemma

An open set $U \subset \mathbb{R}^n$ is *star-shaped with respect to* $p \in U$ if for every $x \in U$ the segment $[p, x]$ lies in U.

Theorem 4.4 (Poincaré lemma)**.** *Let $U \subset \mathbb{R}^n$ be star-shaped. Then every closed k-form on U with $k \geq 1$ is exact, i.e. $H^k(U) = 0$ for $k \geq 1$ (and $H^0(U) = \mathbb{R}$, U being connected).*

Proof. After translating, assume U is star-shaped with respect to 0. We construct an explicit operator $P : \Omega^k(U) \to \Omega^{k-1}(U)$ (for $k \geq 1$, with $P = 0$ on functions) such that

$$dP + Pd = \mathrm{id} \qquad \text{on } \Omega^k(U),\ k \geq 1, \tag{4.7}$$

and $Pdf = f - f(0)$ on functions. Then every closed form ω of degree $k \geq 1$ satisfies $\omega = dP\omega + Pd\omega = d(P\omega)$, so ω is exact, and the claim follows.

For $\omega = f_I\, dx^I \in \Omega^k(U)$, with $I = (i_1 < \cdots < i_k)$, set

$$(P\omega)(x) = \sum_{\alpha=1}^{k} (-1)^{\alpha-1} x^{i_\alpha} \Big(\int_0^1 t^{k-1} f_I(tx)\, dt \Big) dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_\alpha}} \wedge \cdots \wedge dx^{i_k}. \tag{4.8}$$

It remains to verify (4.7). It suffices, by linearity, to treat $\omega = f\, dx^I$ with $k \geq 1$, and we compute the two terms separately. Write $\hat{I}_\alpha$ for the multi-index with i_α deleted and abbreviate $A_\alpha(x) = \int_0^1 t^{k-1} f(tx)\, dt$.

The term $dP\omega$. From (4.8),

$$dP\omega = \sum_\alpha (-1)^{\alpha-1} \Big[A_\alpha\, dx^{i_\alpha} \wedge dx^{\hat{I}_\alpha} + x^{i_\alpha} \sum_j \frac{\partial A_\alpha}{\partial x^j}\, dx^j \wedge dx^{\hat{I}_\alpha} \Big].$$

Since $dx^{i_\alpha} \wedge dx^{\hat{I}_\alpha} = (-1)^{\alpha-1} dx^I$ and $\partial A_\alpha / \partial x^j = \int_0^1 t^k (\partial_j f)(tx)\, dt$ (differentiating under the integral),

$$dP\omega = k\Big(\int_0^1 t^{k-1} f(tx)\, dt \Big) dx^I + \sum_\alpha \sum_j (-1)^{\alpha-1} x^{i_\alpha} \Big(\int_0^1 t^k \frac{\partial f}{\partial x^j}(tx)\, dt \Big) dx^j \wedge dx^{\hat{I}_\alpha}. \tag{4.9}$$

The term $Pd\omega$. First $d\omega = \sum_j \partial_j f\, dx^j \wedge dx^I$. When j is inserted into I at position β (i.e. $i_{\beta-1} < j < i_\beta$, with the obvious conventions at the ends), the rule $dx^j \wedge dx^I = (-1)^{\beta-1} dx^J$, $J = I \cup \{j\}$ sorted, applies. Applying (4.8) to each summand and separating the inserted index ($\alpha = \beta$, sign $(-1)^{\beta-1}$, factor x^j) from the remaining indices ($\alpha \neq \beta$, sign $(-1)^{\alpha-1}$ for $\alpha < \beta$ and $(-1)^\alpha$ for $\alpha > \beta$, since the insertion shifts i_α's position) gives

$$Pd\omega = \sum_j x^j \Big(\int_0^1 t^k \frac{\partial f}{\partial x^j}(tx)\, dt \Big) dx^I - \sum_\alpha \sum_j (-1)^{\alpha-1} x^{i_\alpha} \Big(\int_0^1 t^k \frac{\partial f}{\partial x^j}(tx)\, dt \Big) dx^j \wedge dx^{\hat{I}_\alpha}, \tag{4.10}$$

the sign $(-1)^\alpha$ for $\alpha > \beta$ being rewritten as $-(-1)^{\alpha-1}$ after moving dx^j past dx^{i_α}.

Combining. The double sums in (4.9) and (4.10) cancel, and

$$\begin{aligned} (dP + Pd)\omega &= \Big[k \int_0^1 t^{k-1} f(tx)\, dt + \sum_j x^j \int_0^1 t^k \frac{\partial f}{\partial x^j}(tx)\, dt \Big] dx^I \\ &= \Big[\int_0^1 \frac{d}{dt}\big(t^k f(tx) \big) dt \Big] dx^I = f(x)\, dx^I, \end{aligned} \tag{4.11}$$

since $k \geq 1$ forces $t^k f(tx) \to 0$ as $t \to 0$. This is (4.7). For $k = 0$, $dPf = 0$ and $Pdf = \sum_j x^j \int_0^1 \partial_j f(tx)\, dt = \int_0^1 \frac{d}{dt} f(tx)\, dt = f(x) - f(0)$. □

Remark 4.5. Two comments. First, the lemma is sharp: $H^1(\mathbb{R}^2 \setminus \{0\}) \neq 0$ (Section 4.4) even though the punctured plane is locally star-shaped, so the global obstruction is real. Second, in vector-calculus language the lemma contains the familiar facts "a curl-free vector field on a star-shaped domain is a gradient" and "a divergence-free field is a curl": Exercise 2.12 translates the dictionary.

4.3 Mayer–Vietoris

The Mayer–Vietoris sequence computes the cohomology of $M = U \cup V$ from that of U, V and $U \cap V$. We first record the algebraic lemma behind every long exact sequence.

Lemma 4.6 (Zig-zag lemma). *Let $0 \to A^\bullet \xrightarrow{i} B^\bullet \xrightarrow{j} C^\bullet \to 0$ be a short exact sequence of cochain complexes. Then there is a long exact sequence*

$$\cdots \longrightarrow H^k(A) \xrightarrow{i_*} H^k(B) \xrightarrow{j_*} H^k(C) \xrightarrow{\delta} H^{k+1}(A) \xrightarrow{i_*} H^{k+1}(B) \longrightarrow \cdots \tag{4.12}$$

where $\delta[c] = [a]$ for any choice of $b \in B^k$ with $j(b) = c$ and the unique $a \in A^{k+1}$ with $i(a) = db$.

Proof. If c is a cocycle and $j(b) = c$, then $j(db) = d(jb) = dc = 0$, so $db \in \ker j = \operatorname{im} i$ by exactness, and a is unique by injectivity of i. Also $i(da) = d(ia) = d^2 b = 0$, so $da = 0$: a is a cocycle.

Well-definedness. If b' is another lift of c, then $j(b - b') = 0$, so $b - b' = i(a_0)$ for some $a_0 \in A^k$; the corresponding a' satisfies $i(a') = db' = db - d(ia_0) = i(a - da_0)$, hence $a' = a - da_0$ and $[a'] = [a]$. If $c' = c + d\tilde{c}$, lift $\tilde{c}$ to $\tilde{b}$ and use $b + d\tilde{b}$ as a lift of c': its a equals $a + d(\cdot)$ with $i(a + d\cdot) = db + d^2\tilde{b} = db$, so the class is unchanged.

Exactness. We verify the three representative positions; the remaining ones repeat the same arguments with shifted indices. *At $H^k(C)$*: $\delta j_* = 0$, since for $c = j(b)$ with b a cocycle one may take the lift b itself, and then $db = 0$ forces $a = 0$ in the definition of δ. Conversely, if $\delta[c] = 0$, then for a lift b the cocycle a with $i(a) = db$ is a coboundary, $a = da_0$; then $b - i(a_0)$ is a cocycle ($d(b - ia_0) = db - i(da_0) = i(a) - i(a) = 0$) and $j(b - ia_0) = c$, so $c = j_*[b - ia_0]$. *At $H^k(B)$*: $j_* i_* = 0$ since $ji = 0$; if $j_*[b] = 0$, then $j(b) = dc$ for some $c \in C^{k-1}$; lifting c to $\tilde{b} \in B^{k-1}$ gives $j(b - d\tilde{b}) = dc - dc = 0$, so $b - d\tilde{b} = i(a)$ for some cocycle a, and $[b] = i_*[a]$. *At $H^k(A)$*: $i_*\delta = 0$ because $i(a) = db$ is exact in B; if $i_*[a] = 0$, then $i(a) = db$ for some $b \in B^{k-1}$, and $j(b) \in C^{k-1}$ is a cocycle ($dj(b) = j(db) = j(i(a)) = 0$) with $\delta[j(b)] = [a]$ by construction. ☐

Theorem 4.7 (Mayer–Vietoris). *Let $U, V \subset M$ be open with $M = U \cup V$, and write i_U, i_V and $i_{U\cap V}$ for the inclusions. Then there is a long exact sequence*

$$\cdots \longrightarrow H^k(M) \xrightarrow{(i_U^*, i_V^*)} H^k(U) \oplus H^k(V) \xrightarrow{\text{diff}} H^k(U \cap V) \xrightarrow{\delta} H^{k+1}(M) \longrightarrow \cdots, \tag{4.13}$$

where $\operatorname{diff}(\alpha, \beta) = \alpha|_{U\cap V} - \beta|_{U\cap V}$. *The connecting homomorphism is*

$$\delta[\omega] = [\, d\rho_V \wedge \omega\,], \qquad \omega \in Z^k(U \cap V), \tag{4.14}$$

where $\rho_U + \rho_V = 1$ is a partition of unity subordinate to $\{U, V\}$; the form $d\rho_V \wedge \omega$, supported in $U \cap V$, is extended by zero to M.

Proof. The theorem follows from Lemma 4.6 applied to the short exact sequence of complexes

$$0 \longrightarrow \Omega^k(M) \xrightarrow{(i_U^*, i_V^*)} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{i^*_{U\cap V,U} - i^*_{U\cap V,V}} \Omega^k(U \cap V) \longrightarrow 0, \tag{4.15}$$

so it remains to justify exactness of (4.15). Injectivity of the first map is clear (a form vanishing on both U and V vanishes). If $(\alpha, \beta) \in \Omega^k(U) \oplus \Omega^k(V)$ agree on $U \cap V$, then they glue to a global form ω with $\omega|_U = \alpha$, $\omega|_V = \beta$; hence ker(difference) = im(restriction). For surjectivity of the difference map, take $\gamma \in \Omega^k(U \cap V)$: since $\operatorname{supp} \rho_V \cap U$ is closed in U and contained in $U \cap V$, the form $\rho_V \gamma$ extends by zero to a smooth form on U, and similarly $\rho_U \gamma$ extends to V. Then

$$\rho_V \gamma\big|_{U\cap V} - \big(-\rho_U \gamma\big|_{U\cap V}\big) = (\rho_V + \rho_U)\gamma = \gamma,$$

so $(\rho_V \gamma, -\rho_U \gamma)$ is a preimage of γ.

Now derive (4.14). For a closed $\gamma \in \Omega^k(U \cap V)$ use the lift $(\rho_V \gamma, -\rho_U \gamma)$:

$$d(\rho_V \gamma) = d\rho_V \wedge \gamma + \rho_V d\gamma = d\rho_V \wedge \gamma, \qquad d(-\rho_U \gamma) = -d\rho_U \wedge \gamma = d\rho_V \wedge \gamma,$$

using $d\gamma = 0$ and $d\rho_U + d\rho_V = 0$. The two forms agree on $U \cap V$ and glue to the global closed form $d\rho_V \wedge \gamma$, which is the representative required by Lemma 4.6. □

For completeness we verify the three nontrivial exactness statements of (4.13) directly, since the explicit shape of everything involved will be used in the computations.

Proposition 4.8 (Exactness of the Mayer–Vietoris sequence)**.** *In the sequence* (4.13)*:*

(i) $\ker\big((i_U^*, i_V^*) : H^k(M) \to H^k(U) \oplus H^k(V)\big) = \operatorname{im}\big(\delta : H^{k-1}(U \cap V) \to H^k(M)\big)$*;*

(ii) $\ker\big(H^k(U) \oplus H^k(V) \to H^k(U \cap V)\big) = \operatorname{im}\big(H^k(M) \to H^k(U) \oplus H^k(V)\big)$*;*

(iii) $\ker\big(\delta : H^k(U \cap V) \to H^{k+1}(M)\big) = \operatorname{im}\big(H^k(U) \oplus H^k(V) \to H^k(U \cap V)\big)$*.*

Proof. (i) *First inclusion.* If $\gamma \in Z^{k-1}(U \cap V)$, then $d\rho_V \wedge \gamma|_U = d(\rho_V \gamma)$ is exact on U and, writing $d\rho_V = -d\rho_U$, also $d\rho_V \wedge \gamma|_V = -d\rho_U \wedge \gamma|_V = d(-\rho_U \gamma)$ is exact on V; hence $(i_U^*, i_V^*)\delta[\gamma] = 0$. *Second inclusion.* Suppose $\omega \in Z^k(M)$ with $\omega|_U = d\alpha$, $\omega|_V = d\beta$. Then $\gamma := \alpha|_{U\cap V} - \beta|_{U\cap V}$ is closed on $U \cap V$ (both differentials restrict to ω), and we claim $\delta[\gamma] = [\omega]$. Indeed $(\alpha - \rho_V \gamma,\ \beta + \rho_U \gamma) \in \Omega^{k-1}(U) \oplus \Omega^{k-1}(V)$ agree on the overlap, because there $(\alpha - \rho_V \gamma) - (\beta + \rho_U \gamma) = \gamma - (\rho_U + \rho_V)\gamma = 0$; they glue to $\eta \in \Omega^{k-1}(M)$ with $d\eta|_U = d\alpha - d(\rho_V \gamma) = \omega|_U - d\rho_V \wedge \gamma$ and $d\eta|_V = d\beta + d(\rho_U \gamma) = \omega|_V + d\rho_U \wedge \gamma = \omega|_V - d\rho_V \wedge \gamma$, so $d\eta = \omega - d\rho_V \wedge \gamma$ and $[\omega] = \delta[\gamma]$.

(ii) This is exactness of (4.15) restricted to closed forms, which was proved above.

(iii) *First inclusion.* If $\gamma = \alpha|_{U\cap V} - \beta|_{U\cap V}$ with $\alpha \in Z^k(U)$, $\beta \in Z^k(V)$, then

$$d\rho_V \wedge \gamma = d\rho_V \wedge \alpha - d\rho_V \wedge \beta = d(\rho_V \alpha) - d(\rho_V \beta),$$

where $\rho_V \alpha$ and $\rho_V \beta$ extend by zero to global forms; hence $\delta[\gamma] = 0$. *Second inclusion.* Suppose $\delta[\gamma] = 0$, i.e. $d\rho_V \wedge \gamma = d\eta$ with $\eta \in \Omega^k(M)$ (note the degree: $\gamma \in \Omega^k(U \cap V)$, so $d\rho_V \wedge \gamma$ and $d\eta$ have degree $k+1$). Set

$$\alpha := \rho_V \gamma - \eta|_U \in \Omega^k(U), \qquad \beta := -\rho_U \gamma - \eta|_V \in \Omega^k(V).$$

Then $d\alpha = d\rho_V \wedge \gamma - d\eta|_U = 0$ and $d\beta = d\rho_V \wedge \gamma - d\eta|_V = 0$ (using $d\gamma = 0$ and $d\rho_U = -d\rho_V$), so α, β are closed, and on $U \cap V$

$$\alpha - \beta = (\rho_V + \rho_U)\gamma - (\eta|_U - \eta|_V) = \gamma,$$

since η is globally defined. Thus $[\gamma]$ lies in the image of the difference map. □

Remark 4.9. The sign in (4.14) reflects our choice of lift $(\rho_V \gamma, -\rho_U \gamma)$; choosing $(-\rho_U \gamma, \rho_V \gamma)$ reverses it. What matters is that the same convention is used in all subsequent computations; the factor-of-two bookkeeping in Section 4.4 is a good check that the convention is being applied consistently.

4.4 First computations: S^1 and the punctured plane

Proposition 4.10. *$H^0(S^1) = \mathbb{R}$, $H^1(S^1) = \mathbb{R}$, generated by $[d\theta/2\pi]$, and $H^k(S^1) = 0$ for $k \geq 2$. Here θ is any local angular coordinate; $d\theta$ is globally defined even though θ is not.*

Proof. Cover S^1 by the two open arcs $U = S^1 \setminus \{(1, 0)\}$ and $V = S^1 \setminus \{(-1, 0)\}$, each diffeomorphic to $\mathbb{R}$ (via a stereographic-type projection) and hence contractible. Their intersection $U \cap V$ is the disjoint union of two open arcs, so $H^0(U \cap V) = \mathbb{R} \oplus \mathbb{R}$ and $H^k(U \cap V) = 0$ for $k \geq 1$. The Mayer–Vietoris sequence (4.13) reads, in low degrees,

$$0 \longrightarrow H^0(S^1) \longrightarrow \mathbb{R} \oplus \mathbb{R} \longrightarrow \mathbb{R} \oplus \mathbb{R} \longrightarrow H^1(S^1) \longrightarrow 0,$$

and $H^k(S^1) = 0$ for $k \geq 2$, since the flanking terms vanish. The map $\mathbb{R} \oplus \mathbb{R} \to \mathbb{R} \oplus \mathbb{R}$ sends the pair of constants (c_U, c_V) to $(c_U - c_V, c_U - c_V)$ (the same difference on both components), so its kernel is the diagonal $\mathbb{R}$ and its cokernel is $\mathbb{R}$: $H^0(S^1) = \mathbb{R}$ and $H^1(S^1) = \mathbb{R}$.

It remains to exhibit the generator. With $\theta \in (0, 2\pi)$ the angular coordinate on U, the 1-form $d\theta$ is defined on U; it extends across the missing point to the global form $d\theta = (x\,dy - y\,dx)/(x^2+y^2)$ on S^1 (equivalently, one uses two angular coordinates on U and V differing by a constant, so their differentials agree). It is closed, and $\int_{S^1} d\theta/2\pi = 1$ by direct computation, so $[d\theta/2\pi] \neq 0$ by Stokes' theorem (Theorem 2.10): if $d\theta/2\pi = df$ then the integral would vanish. Hence $H^1(S^1) = \mathbb{R}\,[d\theta/2\pi]$. □

Proposition 4.11. *For $M = \mathbb{R}^2 \setminus \{0\}$:*

$$H^0(M) = \mathbb{R}, \qquad H^1(M) = \mathbb{R} \text{ generated by } \big[d\theta/2\pi\big], \qquad H^k(M) = 0 \text{ for } k \geq 2.$$

Proof. Let $U = \mathbb{R}^2 \setminus \{(-\infty, 0] \times \{0\}\}$ and $V = \mathbb{R}^2 \setminus \{[0, \infty) \times \{0\}\}$ (the plane with the negative, respectively positive, real ray removed). Both U and V are star-shaped with respect to 1 and -1 respectively: for $z \in U$ the segment $[1, z]$ meets the negative ray only if it crosses the real axis at a negative point, but the only point of the segment with imaginary part 0 is 1 itself unless z is real, in which case $z > 0$ keeps the whole segment positive. Hence $H^k(U) = H^k(V) = 0$ for $k \geq 1$ by the Poincaré lemma, and $U \cap V = \mathbb{R}^2 \setminus \mathbb{R}$ is the disjoint union of the upper and lower half-planes, so $H^0(U \cap V) = \mathbb{R} \oplus \mathbb{R}$ and $H^k(U \cap V) = 0$ for $k \geq 1$. Mayer–Vietoris gives

$$0 \to H^0(M) \to \mathbb{R} \oplus \mathbb{R} \to \mathbb{R} \oplus \mathbb{R} \to H^1(M) \to 0$$

and $H^k(M) = 0$ for $k \geq 2$; as in Proposition 4.10 the middle map is the difference map with image the diagonal, so $H^0(M) = \mathbb{R}$, $H^1(M) = \mathbb{R}$.

For the generator, let $f \in C^\infty(U \cap V)$ be 1 on the upper half-plane and 0 on the lower one, and choose the partition of unity with $\rho_V = \theta/\pi$ in the angular variable near the positive axis (a smooth choice, constant near the axes). Then $\delta[f] = [d\rho_V \wedge f] = [d\theta/\pi]$ on the upper half-plane, extended by zero elsewhere; in polar coordinates this form is $d\theta/\pi$ everywhere, hence $\delta[f] = 2\,[d\theta/2\pi]$. Since δ is onto $H^1(M) \cong \mathbb{R}$, the class $[d\theta/2\pi]$, whose integral over the unit circle is 1, generates $H^1(M)$. □

Remark 4.12. The factor of two in the last paragraph is not a mistake but a useful warning: connecting homomorphisms depend on the choice of partition of unity, and only the resulting *isomorphism* is canonical up to the identification. The clean statement is: the class with $\int_{S^1} [d\theta/2\pi] = 1$ generates.

Exercise 4.13. Compute $H^*(S^1 \times S^1)$ by two applications of the Mayer–Vietoris sequence: first cut the torus into two cylinders $U = S^1 \times (\text{arc})$, $V = S^1 \times (\text{arc})$, and use Proposition 4.10 plus the fact (proved in Section 6.4) that the pullback of the projection kills only the fibre directions. Answer: $H^0 = \mathbb{R}$, $H^1 = \mathbb{R}^2$, $H^2 = \mathbb{R}$.

4.5 The generalized Mayer–Vietoris sequence

We close the chapter with the finite-cover version of the sequence, which is the formal engine behind the de Rham theorem in Chapter 5.

Theorem 4.14 (Generalized Mayer–Vietoris, Bott–Tu)**.** *Let $\mathcal{U} = \{U_0, \ldots, U_m\}$ be a finite open cover of M, and write $U_{\alpha_0 \ldots \alpha_p} = U_{\alpha_0} \cap \cdots \cap U_{\alpha_p}$. Then the sequence*

$$0 \longrightarrow \Omega^k(M) \xrightarrow{r} \prod_{\alpha_0} \Omega^k(U_{\alpha_0}) \xrightarrow{\delta} \prod_{\alpha_0<\alpha_1} \Omega^k(U_{\alpha_0\alpha_1}) \xrightarrow{\delta} \prod_{\alpha_0<\alpha_1<\alpha_2} \Omega^k(U_{\alpha_0\alpha_1\alpha_2}) \xrightarrow{\delta} \cdots, \quad (4.16)$$

where r restricts a form to each U_α and $(\delta\omega)_{\alpha_0\ldots\alpha_{p+1}} = \sum_{j=0}^{p+1}(-1)^j \, \omega_{\alpha_0\ldots\widehat{\alpha_j}\ldots\alpha_{p+1}}\big|_{U_{\alpha_0\cdots\alpha_{p+1}}}$, is exact for every k.

The proof is a collating argument with a partition of unity; it is given in Section 5.2, where the sequence reappears as the rows of the Čech–de Rham double complex. One consequence, which we use repeatedly, is that for an open cover with all intersections H^k-acyclic one can compute $H^k(M)$ by the same telescoping procedure that produced the Mayer–Vietoris sequence for two open sets; this is made precise in Chapter 5.

5 The de Rham theorem

We now assemble the local pieces into the global statement: de Rham cohomology is topological. The strategy, due to A. Weil and presented in Bott–Tu [1, §8], runs through Čech cohomology: (i) the generalized Mayer–Vietoris sequence makes the de Rham complex of M quasi-isomorphic to a Čech complex built from a "good" cover; (ii) for good covers that Čech complex computes singular cohomology. Both steps are proved in detail below.

5.1 Good covers

Definition 5.1. A *good cover* of a manifold M is an open cover $\mathcal{U} = \{U_\alpha\}$ such that every nonempty finite intersection $U_{\alpha_0} \cap \cdots \cap U_{\alpha_p}$ is diffeomorphic to $\mathbb{R}^n$. The cover is *finite* if it has finitely many elements.

Proposition 5.2 (Existence of good covers)**.** *Every compact manifold admits a finite good cover; every manifold admits a good cover (possibly infinite).*

Proof. For compact M, fix a Riemannian metric. For every $p \in M$ let U_p be a geodesically convex ball around p (Lee [2, Thm. 6.17]), i.e. a diffeomorphic image of a convex ball in $\mathbb{R}^n$ with the property that any two points are joined by a unique minimizing geodesic lying in U_p. Finite intersections of geodesically convex sets are geodesically convex (the uniqueness of the minimizing segment is inherited), and a geodesically convex open set is diffeomorphic to a convex open set in $\mathbb{R}^n$ via the exponential map at any of its points. Compactness gives a finite subcover. For general M the same construction with a locally finite refinement yields an infinite good cover; see also Bott–Tu [1, §5]. □

Remark 5.3. Every good cover is "H^*-acyclic" in the following strong sense: by the Poincaré lemma (Theorem 4.4) and Proposition 3.6,

$$H^k(U_{\alpha_0\ldots\alpha_p}) = \begin{cases} \mathbb{R} & k = 0, \\ 0 & k \geq 1, \end{cases} \quad (5.1)$$

because each intersection is contractible and connected. This single property is all the argument below needs.

5.2 The generalized Mayer–Vietoris sequence

Proof of Theorem 4.14. We must show exactness of (4.16) at every position $p \geq 1$ (the position $p = 0$ is the injectivity of r, which is clear). Choose a partition of unity $\{\rho_\alpha\}_{\alpha=0}^m$ subordinate to $\mathcal{U}$ and define the *collating operators*

$$(K\omega)_{\alpha_0\ldots\alpha_{p-1}} = \sum_\beta \rho_\beta\, \omega_{\beta\alpha_0\ldots\alpha_{p-1}}, \tag{5.2}$$

where each summand is understood to be extended by zero outside $U_\beta \cap U_{\alpha_0\ldots\alpha_{p-1}}$; this is legitimate because ρ_β vanishes near the boundary of that intersection inside $U_{\alpha_0\ldots\alpha_{p-1}}$ (it is supported in U_β). The right-hand side is then a smooth form on $U_{\alpha_0\ldots\alpha_{p-1}}$. We claim that

$$\delta K + K\delta = \mathrm{id}\,. \tag{5.3}$$

Compute the $(\alpha_0 \ldots \alpha_p)$-component of $K\delta\omega$:

$$\begin{aligned}(K\delta\omega)_{\alpha_0\ldots\alpha_p} &= \sum_\beta \rho_\beta(\delta\omega)_{\beta\alpha_0\ldots\alpha_p} \\ &= \sum_\beta \rho_\beta \Big[\omega_{\alpha_0\ldots\alpha_p} + \sum_{j=0}^{p}(-1)^{j+1}\omega_{\beta\alpha_0\ldots\widehat{\alpha_j}\ldots\alpha_p}\Big] \\ &= \omega_{\alpha_0\ldots\alpha_p} - \sum_j (-1)^j \sum_\beta \rho_\beta\, \omega_{\beta\alpha_0\ldots\widehat{\alpha_j}\ldots\alpha_p},\end{aligned} \tag{5.4}$$

using $\sum_\beta \rho_\beta = 1$ (all restrictions being taken on the common domain $U_{\alpha_0\ldots\alpha_p}$). The double sum is exactly $(\delta K\omega)_{\alpha_0\ldots\alpha_p}$: expanding $\delta(K\omega)$ gives $\sum_j(-1)^j(K\omega)_{\alpha_0\ldots\widehat{\alpha_j}\ldots\alpha_p} = \sum_j(-1)^j\sum_\beta \rho_\beta\,\omega_{\beta\alpha_0\ldots\widehat{\alpha_j}\ldots\alpha_p}$. Hence (5.3). Now if $\delta\omega = 0$, then $\omega = \delta(K\omega) + K\delta\omega = \delta(K\omega)$ is a coboundary, so the sequence is exact at every position $p \geq 1$. □

Remark 5.4. The operator (5.2) is the Čech analogue of the cylinder operator (4.3) and of the Poincaré operator (4.8): all three are "integration against a resolution of the identity", and all three prove exactness by exhibiting a contracting homotopy. The same formula with coefficients in any sheaf of forms proves that the sheaves $\underline{\Omega}^q$ are *soft* (or, for the present purposes, simply that $H^p_\delta(K^{*,q}) = 0$ for $p \geq 1$).

5.3 The Čech–de Rham double complex

Fix a finite good cover $\mathcal{U} = \{U_0, \ldots, U_m\}$ of M. Form the double complex

$$K^{p,q} = \prod_{\alpha_0<\cdots<\alpha_p} \Omega^q(U_{\alpha_0\ldots\alpha_p}), \tag{5.5}$$

with the Čech differential $\delta : K^{p,q} \to K^{p+1,q}$ of (4.16) as horizontal differential, the exterior derivative on each factor as vertical differential, and the sign-adjusted total differential

$$D = \delta + (-1)^p d \qquad \text{on } K^{p,q}. \tag{5.6}$$

One checks $D^2 = 0$: $\delta^2 = 0$ (the usual alternating-sum computation, identical to the one for the de Rham complex with d replaced by δ), $d^2 = 0$, and the cross terms cancel since δ and d commute and the sign $(-1)^p$ produces -1 exactly when the two differentials pass each other. We write $H^k_D(K)$ for the cohomology of the total complex $(\bigoplus_{p+q=k} K^{p,q}, D)$.

The two filtrations of K, by columns (p) and by rows (q), give two descriptions of $H^k_D(K)$, and comparing them proves the theorem. The abstract fact needed is the following edge comparison, whose proof is the explicit zigzag that spectral sequences abbreviate.

Lemma 5.5 (Edge comparison for bounded double complexes)**.** *Let (K, d, δ) be a double complex with $K^{p,q} = 0$ unless $0 \le p \le m$, $0 \le q \le n$, and total differential $D = \delta + (-1)^p d$.*

(a) If $H_d^q(K^{p,*}) = 0$ for all $q \ge 1$ and all p, then $H_D^k(K) \cong H^k(B^*, \bar\delta)$, where $B^p = \ker\big(d : K^{p,0} \to K^{p,1}\big)$ and $\bar\delta$ is induced by δ.

(b) If $H_\delta^p(K^{*,q}) = 0$ for all $p \ge 1$ and all q, then $H_D^k(K) \cong H^k(A^*, \bar d)$, where $A^q = \ker\big(\delta : K^{0,q} \to K^{1,q}\big)$ and $\bar d$ is induced by d.

Proof. The two statements are transposes of one another ($p \leftrightarrow q$, $\delta \leftrightarrow d$), so we prove (a).

Construction of the map. Let $x \in \bigoplus_{p+q=k} K^{p,q}$ be a D-cocycle, with components $x^{p,q} \in K^{p,q}$. We modify x within its cohomology class, killing the components $x^{0,k}, x^{1,k-1}, \dots, x^{k-1,1}$ one by one. The cocycle condition $(Dx)^{p,q} = \delta(x^{p-1,q}) + (-1)^p d(x^{p,q-1}) = 0$ for all (p,q) with $p + q = k + 1$ implies, at $(p,q) = (0, k+1)$, $d(x^{0,k}) = 0$. Since column 0 is d-acyclic in positive degrees, $x^{0,k} = d(z^{0,k-1})$ for some $z^{0,k-1} \in K^{0,k-1}$. Replace x by $x - D\tilde z$, where $\tilde z \in \bigoplus_{p+q=k-1} K^{p,q}$ is z in component $(0, k-1)$ and zero elsewhere; the new cocycle (still called x) has $x^{0,k} = 0$. Now the cocycle condition at $(1,k)$ reads $\delta(x^{0,k}) + (-1)^1 d(x^{1,k-1}) = 0$, hence $d(x^{1,k-1}) = 0$; column 1 is d-acyclic in positive degrees, so $x^{1,k-1} = d(z^{1,k-2})$, and subtracting $D\tilde z$ kills $x^{1,k-1}$. Proceeding in this way we arrive at a cocycle x with $x^{p,q} = 0$ for all $q \ge 1$, i.e. $x = x^{k,0}$. The cocycle condition at $(k,1)$ then gives $d(x^{k,0}) = 0$ (the δ-term involves the killed component $x^{k-1,1}$), and the condition at $(k+1, 0)$ is vacuous; thus $x^{k,0} \in B^k$. Moreover $\delta(x^{k,0}) = (Dx)^{k+1,0} = 0$, so $x^{k,0}$ is a $\bar\delta$-cocycle. Set $\Phi([x]) = [x^{k,0}]$.

Well-definedness. The auxiliary forms $z^{p,q}$ are determined up to d-closed elements; replacing z by $z + u$ with $du = 0$ changes the subsequent cocycle by $D\tilde u$, whose effect on the final component $x^{k,0}$ is to add $\delta(w)$ for some $w \in B^{k-1}$: indeed $(D\tilde u)^{p+1,q-1} = \delta(u^{p,q-1})$ pushes the correction one column to the right, and iterating reaches $x^{k,0}$ as a $\bar\delta$-coboundary. Hence the class $[x^{k,0}]$ is unchanged. If $x = Dy$ is a coboundary, then the very first step already exhibits $x^{0,k} = d(y^{0,k-1})$ (choosing $z = y$), and the construction terminates with $x^{k,0} = \delta(y^{k-1,0})$, a $\bar\delta$-coboundary in B^*; so Φ kills exact classes.

The inverse. Send $[b] \in H^k(B^*, \bar\delta)$ to the class of b, viewed as an element of $K^{k,0} \subset \bigoplus K^{p,q}$: since $db = 0$ and $\delta b = 0$, indeed $Db = 0$. If b is a $\bar\delta$-coboundary, $b = \delta b'$ with $b' \in B^{k-1}$, then $b = Db'$, so the map is well-defined on cohomology. The two constructions are visibly inverse: Φ does not move a cocycle supported in $K^{k,0}$, and any cocycle is modified only by D-boundaries. ☐

Theorem 5.6 (De Rham theorem for finite good covers)**.** *Let M admit a finite good cover $\mathcal{U} = \{U_0, \dots, U_m\}$. Then there are canonical isomorphisms*

$$H^k_{\mathrm{dR}}(M) \;\cong\; \check H^k(\mathcal{U}; \mathbb{R}), \tag{5.7}$$

where $\check H^*(\mathcal{U}; \mathbb{R})$ is the cohomology of the Čech complex

$$C^p(\mathcal{U}; \mathbb{R}) = \prod_{\alpha_0 < \cdots < \alpha_p} \mathbb{R}, \qquad (\bar\delta c)_{\alpha_0 \dots \alpha_{p+1}} = \sum_{j=0}^{p+1} (-1)^j c_{\alpha_0 \dots \widehat{\alpha_j} \dots \alpha_{p+1}}.$$

Proof. Compute $H_D^k(K)$ in two ways. *First filtration.* Since $H_d^q(K^{p,*}) = \prod_{\alpha_0 < \cdots < \alpha_p} H^q(U_{\alpha_0 \dots \alpha_p})$, the good-cover acyclicity (5.1) gives $H_d^q(K^{p,*}) = 0$ for $q \ge 1$ and $B^p = \prod_{\alpha_0 < \cdots < \alpha_p} \mathbb{R}$ (a closed function on a connected contractible intersection is constant). Lemma 5.5(a) applies, and $H_D^k(K) \cong \check H^k(\mathcal{U}; \mathbb{R})$.

Second filtration. The generalized Mayer–Vietoris theorem (Theorem 4.14) says precisely that $H_\delta^p(K^{*,q}) = 0$ for $p \ge 1$, all q, with $A^q = \ker\big(\delta : K^{0,q} \to K^{1,q}\big) = \Omega^q(M)$ (a collection of forms on the U_α's whose differences vanish on the overlaps is exactly a global form). Lemma 5.5(b) applies, and $H_D^k(K) \cong H^k_{\mathrm{dR}}(M)$.

Comparing the two descriptions proves (5.7). ☐

5.4 Comparison with singular cohomology

The remaining step is standard: for a good cover, the Čech complex computes singular cohomology.

Theorem 5.7 (Nerve theorem for good covers)**.** *If $\mathcal{U}$ is a good cover of M, then $\check{H}^k(\mathcal{U};\mathbb{R}) \cong H^k_{\mathrm{sing}}(M;\mathbb{R})$ for all k, canonically.*

Sketch. For finite $\mathcal{U}$ the proof is classical: choose a Riemannian metric on M. The nerve $N(\mathcal{U})$ of the cover is the simplicial complex with one vertex per U_α and one p-simplex per nonempty intersection $U_{\alpha_0\ldots\alpha_p}$. Because finite intersections are geodesically convex, one constructs a homotopy equivalence $|N(\mathcal{U})| \simeq M$ by geodesic barycentric subdivision: map each open simplex of $|N(\mathcal{U})|$ into the corresponding intersection using a smooth partition of unity subordinate to $\mathcal{U}$ (the inverse map is the barycentric coordinate map, well-defined and continuous thanks to the local finiteness of the cover). Then $\check{H}^k(\mathcal{U};\mathbb{R}) \cong H^k(|N(\mathcal{U})|;\mathbb{R}) \cong H^k_{\mathrm{sing}}(M;\mathbb{R})$; the first isomorphism is standard (see Hatcher [6, §3.2, 4G] or Bott–Tu [1, §8]). For infinite good covers one takes a direct limit over finite subcovers; details are again in Bott–Tu [1, §8] and Warner [3, §5]. □

Theorem 5.8 (de Rham, 1931; Weil, 1952)**.** *For every smooth manifold M and every $k \geq 0$ the integration pairing induces an isomorphism*

$$H^k_{\mathrm{dR}}(M) \xrightarrow{\cong} H^k_{\mathrm{sing}}(M;\mathbb{R}), \qquad [\omega] \longmapsto \Big([\sigma] \mapsto \int_\sigma \omega\Big). \tag{5.8}$$

Proof for manifolds with finite good covers. Every compact manifold has a finite good cover (Proposition 5.2), so this case covers all compact M. First, the pairing is well-defined: if $\omega = d\eta$ then $\int_\sigma \omega = \int_{\partial\sigma} \eta = 0$ on cycles by Stokes' theorem, and if $\partial\tau = \sigma - \sigma'$ then $\int_\sigma \omega - \int_{\sigma'} \omega = \int_{\partial\tau} \omega = \int_\tau d\omega = 0$. Write $I_M : H^k_{\mathrm{dR}}(M) \to H^k_{\mathrm{sing}}(M;\mathbb{R})$ for the resulting map (with smooth singular chains; smooth and continuous singular homology agree, Lee [2, Ch. 18]).

We prove I_M is an isomorphism by induction on the number m of sets in a finite good cover of M, using the Mayer–Vietoris sequence and the five lemma. The induction needs the following facts, all standard: singular cohomology satisfies homotopy invariance and has a Mayer–Vietoris long exact sequence for any open cover $\{U, V\}$ (Hatcher [6, §3.2]); I is natural for restriction to open subsets; and I commutes with the two connecting homomorphisms, in the sense that

$$\langle I_M \delta_{\mathrm{dR}}[\gamma], [\sigma]\rangle = \langle \delta_{\mathrm{sing}} I_{U\cap V}[\gamma], [\sigma]\rangle \tag{5.9}$$

for $[\gamma] \in H^k(U \cap V)$ and $[\sigma] \in H_{k+1}(M;\mathbb{R})$. Verification of (5.9): by (4.14) the left-hand side is $\int_\sigma d\rho_V \wedge \gamma = \int_{\partial\sigma} \rho_V \gamma$ (Stokes). Subdivide σ (as a smooth chain, up to reparametrization) into $\sigma = \sigma_U + \sigma_V$ with $\operatorname{supp}\sigma_V \subset V$ and $\operatorname{supp}\partial\sigma_V \subset U \cap V$ (use the Lebesgue number of the cover on the standard simplices). Then $\int_{\partial\sigma} \rho_V\gamma = \int_{\partial\sigma_V} \gamma$, because $\rho_V \equiv 1$ on σ_V's interior and $\rho_V\gamma$ is supported in $U \cap V$. The right-hand side of (5.9) is $\langle I_{U\cap V}\gamma, \partial\sigma_V\rangle = \int_{\partial\sigma_V} \gamma$ by the definition of the singular connecting homomorphism (the dual of the homology Mayer–Vietoris map $\partial[\sigma] = [\partial\sigma_V]$). This proves the compatibility.

Base of the induction ($m = 1$): $M = U_0$ is contractible; $H^k_{\mathrm{dR}}(M)$ is $\mathbb{R}$ at $k = 0$ and 0 otherwise (Corollary 4.2), and I sends [1] to the cocycle $\sigma \mapsto \sum_i a_i$ for a 0-chain $\sum_i a_i\sigma_i$, which evaluates to 1 on a point; hence I is an isomorphism.

Induction step. Write $M = U \cup V$ with U covered by $U_0, \ldots, U_{m-1}$ and $V = U_m$. Then U and $U \cap V$ admit good covers with at most $m - 1$ sets ($U_i \cap V$, $i < m$, cover $U \cap V$), so I_U, I_V, $I_{U\cap V}$ are isomorphisms by induction on m, in every degree. The Mayer–Vietoris sequences of de Rham cohomology and singular cohomology (Theorem 4.7 and the standard one) fit into a ladder in which all squares commute by the naturality and (5.9). To apply the five lemma at degree k one needs the vertical map in degree $k + 1$ (the fifth term of the ladder); we therefore prove the statement for all degrees at once by *descending induction on k*, starting from $k = n + 1$, where both H^k and H^k_{sing} vanish for degree reasons (forms of degree exceeding $\dim M$ are zero, and singular cohomology vanishes above the dimension). At each step the five lemma then applies, with the degree-$(k+1)$ term already known, and yields I_M an isomorphism in degree k. □

Remark 5.9. The general case of Theorem 5.8, for manifolds with only infinite good covers, needs a limiting argument over the good cover; the standard treatment is Weil's original proof [7] or the spectral-sequence computation of the Čech–de Rham double complex for arbitrary covers (Bott–Tu [1, §8]). The statements of this chapter used in the sequel—finite-dimensionality for compact manifolds, Poincaré duality—all hold for arbitrary manifolds, but the proofs below are written, without further comment, for the finite-good-cover case, which is the case of interest for compact manifolds.

5.5 First consequences

Corollary 5.10 (Finite-dimensionality and Betti numbers)**.** *If M is compact, then* $\dim H^k(M) < \infty$ *for all k, and the Euler characteristic* $\chi(M) = \sum_k (-1)^k b_k$, $b_k = \dim H^k_{\mathrm{sing}}(M;\mathbb{R})$, *equals* $\sum_k (-1)^k \dim H^k_{\mathrm{dR}}(M)$.

Proof. Choose a finite good cover $U_0, \dots, U_m$ (Proposition 5.2) and induct on m. For $m = 1$ the claim is the Poincaré lemma. For the induction step write $M = U \cup V$ with $U = U_0 \cup \dots \cup U_{m-1}$ and $V = U_m$. The Mayer–Vietoris exact sequence (4.13) exhibits $H^k(M)$ as an extension of a subspace of $H^k(U) \oplus H^k(V)$ by a quotient of $H^k(U \cap V)$: the portion $H^k(U \cap V) \to H^k(M) \to H^k(U) \oplus H^k(V)$ shows that the image of the first map is a quotient of $H^k(U \cap V)$ and that the quotient of $H^k(M)$ by this image embeds into $H^k(U) \oplus H^k(V)$. Both flanking spaces are finite-dimensional by induction (for U and $U \cap V$, whose good covers have at most $m - 1$ sets), so $H^k(M)$ is finite-dimensional. The statement about Euler characteristics follows from Theorem 5.8. □

Remark 5.11. The corollary is where de Rham theory makes contact with classical geometry: $\chi(M)$ can be computed from any Riemannian metric as the integral of the Pfaffian of the curvature (the generalized Gauss–Bonnet–Chern theorem), a computation that runs entirely inside the language of forms. This circle of ideas—curvature representing cohomology—is the starting point of Chern–Weil theory, briefly mentioned in Chapter 8.

6 Compactly supported cohomology, Poincaré duality, Künneth

For noncompact manifolds the natural variant of the de Rham complex is the subcomplex of compactly supported forms. Its cohomology pairs with ordinary de Rham cohomology to give Poincaré duality, the deepest structural fact of the subject, and it is the right setting for the Künneth formula of Section 6.4.

6.1 Definitions and the Mayer–Vietoris sequence

Definition 6.1. Let $\Omega^k_c(M) \subset \Omega^k(M)$ be the space of k-forms with compact support. Since d preserves supports, $(\Omega^*_c(M), d)$ is a subcomplex of the de Rham complex; its cohomology

$$H^k_c(M) = \frac{\ker\big(d : \Omega^k_c \to \Omega^{k+1}_c\big)}{\operatorname{im}\big(d : \Omega^{k-1}_c \to \Omega^k_c\big)} \tag{6.1}$$

is the *compactly supported de Rham cohomology* of M.

Warning 6.2. H^*_c is *not* homotopy invariant: $H^n_c(\mathbb{R}^n) = \mathbb{R}$ while $H^*_c(\mathrm{pt}) = 0$ in positive degrees. It is covariant for proper maps (the pullback of a compactly supported form along a proper map has compact support), and for open inclusions it is covariant via extension by zero: $i_* : \Omega_c^k(U) \to \Omega_c^k(M)$, $\omega \mapsto$ the extension of ω by 0 outside U. We use this extension map constantly.

Proposition 6.3 (Mayer–Vietoris for compact supports)**.** *For open $U, V \subset M$ with $M = U \cup V$ there is a long exact sequence*

$$\cdots \longrightarrow H_c^k(U\cap V) \xrightarrow{((i_U)_*,(i_V)_*)} H_c^k(U)\oplus H_c^k(V) \xrightarrow{(i_U)_*-(i_V)_*} H_c^k(M) \xrightarrow{\delta_c} H_c^{k+1}(U\cap V) \longrightarrow \cdots, \tag{6.2}$$

with $\delta_c[\tau] = [\, d\rho_U \wedge \tau\,]$, where $\rho_U + \rho_V = 1$ is a partition of unity subordinate to $\{U, V\}$.

Proof. The short exact sequence of complexes

$$0 \longrightarrow \Omega_c^k(U \cap V) \xrightarrow{((i_U)_*,(i_V)_*)} \Omega_c^k(U) \oplus \Omega_c^k(V) \xrightarrow{(i_U)_*-(i_V)_*} \Omega_c^k(M) \longrightarrow 0$$

is exact: the first map is clearly injective; if $(i_U)_*\alpha = (i_V)_*\beta$ then the common extension is supported in $U \cap V$, hence is $(i_{U\cap V})_*\gamma$ for $\gamma \in \Omega_c^k(U \cap V)$; and every $\tau \in \Omega_c^k(M)$ is $(i_U)_*(\rho_U\tau) - (i_V)_*(-\rho_V\tau)$ (the two pieces are smooth, with supports in U and V respectively). Lemma 4.6 now produces (6.2); the connecting homomorphism is computed as in Theorem 4.7: lift $\tau \in Z_c^k(M)$ to $(\rho_U\tau, -\rho_V\tau)$, differentiate, and glue: both components give $d\rho_U \wedge \tau$, a closed form supported in $U \cap V$. □

6.2 The compact Poincaré lemma and the suspension isomorphism

The central computation is Bott–Tu's Proposition 4.6, which we prove by integrating along the fibres of the projection $\pi : M \times \mathbb{R} \to M$.

Write coordinates (x, t) on $M \times \mathbb{R}$. Every form $\omega \in \Omega_c^k(M \times \mathbb{R})$ is uniquely $\omega = dt \wedge \alpha + \beta$ where $\alpha \in \Omega^{k-1}$ and $\beta \in \Omega^k$ involve no dt. Define the *fibre integral* (integration along the fibre)

$$\pi_* : \Omega_c^k(M \times \mathbb{R}) \to \Omega_c^{k-1}(M), \qquad \pi_*(dt \wedge \alpha + \beta) = \int_{\mathbb{R}} \alpha(\cdot, t)\, dt. \tag{6.3}$$

The integral converges because α has compact support, and $\pi_*\omega$ has compact support in M. A direct computation gives $d\pi_* = -\pi_* d$: with $d = d_x + dt \wedge \partial_t$,

$$d\omega = dt \wedge (\partial_t\beta - d_x\alpha) + d_x\beta, \qquad \pi_*(d\omega) = \int_{\mathbb{R}} (\partial_t\beta - d_x\alpha)\, dt = -d_x \int_{\mathbb{R}} \alpha\, dt = -d\,\pi_*\omega,$$

the term $\int \partial_t\beta\, dt$ vanishing since β has compact support in t. (The sign $-$ is immaterial for what follows; it can be removed by a degree-dependent sign in the definition of π_*, at the cost of bookkeeping in the statements below.)

Lemma 6.4. *Let $\omega \in Z_c^k(M \times \mathbb{R})$ with $k \geq 1$ and $\pi_*\omega = 0$. Then $\omega = d\gamma$ for some $\gamma \in \Omega_c^{k-1}(M \times \mathbb{R})$.*

Proof. Write $\omega = dt \wedge \alpha + \beta$; the hypothesis is $\int_{\mathbb{R}} \alpha\, dt = 0$. Define

$$\gamma(x, t) = \int_{-\infty}^{t} \alpha(x, s)\, ds. \tag{6.4}$$

Then $\gamma \in \Omega_c^{k-1}(M \times \mathbb{R})$: for $t \ll 0$ the integral is 0, for $t \gg 0$ it is $\int_{\mathbb{R}} \alpha = 0$, and the support in x is contained in the (compact) x-support of α. Moreover $d\gamma = dt \wedge \alpha + d_x\gamma$, so $\omega - d\gamma = \beta - d_x\gamma$ involves no dt. It is closed, and closedness of a form without dt forces $\partial_t(\beta - d_x\gamma) = 0$ (the dt-coefficient of $d(\beta - d_x\gamma)$ is exactly that derivative), so $\beta - d_x\gamma$ is independent of t; being compactly supported in t, it vanishes. Hence $\omega = d\gamma$. □

Theorem 6.5 (Compact Poincaré lemma; suspension)**.** *For every manifold M, the fibre integral induces an isomorphism*

$$\pi_* : H^k_c(M \times \mathbb{R}) \xrightarrow{\cong} H^{k-1}_c(M), \qquad k \geq 1. \tag{6.5}$$

Consequently $H^k_c(\mathbb{R}^n) \cong \mathbb{R}$ for $k = n$ and $H^k_c(\mathbb{R}^n) = 0$ for $k < n$.

Proof. Well-definedness. If $\omega \in Z^k_c(M \times \mathbb{R})$ then $d(\pi_*\omega) = -\pi_* d\omega = 0$, so $\pi_*\omega$ is closed; if $\omega = d\eta$, $\pi_*\omega = \pm d(\pi_*\eta)$. Thus π_* descends to cohomology.

Surjectivity. Fix $e \in C^\infty_c(\mathbb{R})$ with $\int_{\mathbb{R}} e(t)\,dt = 1$. For a closed $\tau \in \Omega^{k-1}_c(M)$ set $\omega = \pi^*\tau \wedge e(t)\,dt \in \Omega^k_c(M \times \mathbb{R})$. Then

$$d\omega = \pi^*(d\tau) \wedge e\,dt + (-1)^{k-1}\pi^*\tau \wedge de \wedge dt = 0,$$

because $d\tau = 0$ and $de = e'(t)\,dt$ forces $de \wedge dt = e'(t)\,dt \wedge dt = 0$. Moreover $\pi^*\tau \wedge e\,dt = (-1)^{k-1} e\,dt \wedge \pi^*\tau$, so $\pi_*\omega = (-1)^{k-1}\tau$. Hence $\pi_*[\omega] = \pm[\tau]$.

Injectivity. Let $\omega \in Z^k_c(M \times \mathbb{R})$ with $\pi_*\omega = d\sigma$ for $\sigma \in \Omega^{k-2}_c(M)$ (for $k = 1$ this forces $\pi_*\omega = 0$, since a compactly supported closed function is 0; the argument then skips to the next step). Put $\gamma_0 = \pi^*\sigma \wedge e\,dt \in \Omega^{k-1}_c(M \times \mathbb{R})$. Then $d\gamma_0 = \pi^*(d\sigma) \wedge e\,dt$, and

$$\pi_*\big(d\gamma_0\big) = \pi_*\big(\pi^*\pi_*\omega \wedge e\,dt\big) = (-1)^{k-1}\pi_*\omega.$$

Hence $\pi_*\big(\omega - (-1)^{k-1}d\gamma_0\big) = 0$, and by Lemma 6.4, $\omega - (-1)^{k-1}d\gamma_0 = d\gamma$ with $\gamma \in \Omega^{k-1}_c(M \times \mathbb{R})$; thus $\omega = d\big((-1)^{k-1}\gamma_0 + \gamma\big)$ is exact in Ω^*_c.

The statement about $\mathbb{R}^n$ follows by iterating (6.5): $H^k_c(\mathbb{R}^n) \cong H^{k-1}_c(\mathbb{R}^{n-1}) \cong \ldots \cong H^{k-n}_c(\mathbb{R}^0)$, and $H^0_c(\mathbb{R}^0) = \mathbb{R}$ while $H^0_c(\mathbb{R}^m) = 0$ for $m \geq 1$ (a closed compactly supported function is constant, hence zero). ☐

Remark 6.6. The class of $\mu = e(x^1)\ldots e(x^n)\,dx^1 \wedge \cdots \wedge dx^n$, with $\int_{\mathbb{R}^n}\mu = 1$, generates $H^n_c(\mathbb{R}^n)$: the proof shows $[\mu]$ corresponds, under the iterated suspension isomorphisms, to $[1] \in H^0_c(\mathbb{R}^0)$.

Example 6.7. (a) If M is compact then $\Omega^k_c(M) = \Omega^k(M)$, so $H^k_c(M) = H^k(M)$; in particular $H^k_c(S^n) = H^k(S^n)$ (computed in Section 7). (b) For the punctured plane $M = \mathbb{R}^n \setminus \{0\}$, $n \geq 2$:

$$H^k_c(M) = \begin{cases} \mathbb{R} & k \in \{1, n\}, \\ 0 & \text{otherwise.} \end{cases}$$

With U, V as in Proposition 4.11, the intersection $U \cap V$ is a union of two half-spaces, hence diffeomorphic to $(\mathbb{R}^{n-1} \times \mathbb{R}) \sqcup (\mathbb{R}^{n-1} \times \mathbb{R})$, and $H^k_c(U \cap V) \cong H^k_c(\mathbb{R}^n) \oplus H^k_c(\mathbb{R}^n)$, i.e. $\mathbb{R} \oplus \mathbb{R}$ for $k = n$ and 0 otherwise. For $k \leq n-2$ the compact Mayer–Vietoris sequence (6.2) gives $H^k_c(M) \cong H^k_c(U \cap V) = 0$ (the flanking terms $H^k_c(U) = H^k_c(\mathbb{R}^n)$ and $H^{k+1}_c(U \cap V)$ vanish for $k \leq n-2$, since $k < n$ and $k+1 < n$). For $k = n-1$, the sequence reads

$$0 \longrightarrow H^{n-1}_c(M) \longrightarrow \mathbb{R} \oplus \mathbb{R} \longrightarrow \mathbb{R} \oplus \mathbb{R} \longrightarrow H^n_c(M) \longrightarrow 0,$$

where the map $\mathbb{R} \oplus \mathbb{R} \to \mathbb{R} \oplus \mathbb{R}$ is extension by zero of the two half-space generators, i.e. $(x, y) \mapsto (x+y, x+y)$ (both extensions carry the same integral, the two half-spaces being oriented consistently by $\mathbb{R}^n$); its kernel and cokernel are both $\mathbb{R}$. Hence $H^{n-1}_c(M) = 0$ and $H^n_c(M) = \mathbb{R}$, and $H^1_c(M) = 0$ for $n \geq 3$ by the $k \leq n-2$ case. For $n = 2$, the same sequence at $k = 1$ gives $H^1_c(M) = \ker\big(\mathbb{R} \oplus \mathbb{R} \xrightarrow{(x,y)\mapsto(x+y,x+y)} \mathbb{R} \oplus \mathbb{R}\big) = \mathbb{R}$ (with $H^2_c(M) = \mathbb{R}$ as before). This completes the computation; note that it pairs, via Poincaré duality below, with $H^*(\mathbb{R}^n \setminus \{0\}) = H^*(S^{n-1})$ computed in Section 4.4 and Proposition 7.1.

6.3 Poincaré duality

Let M be an oriented n-manifold. The wedge product and integration define a pairing

$$H^k(M) \times H_c^{n-k}(M) \longrightarrow \mathbb{R}, \qquad \langle [\omega], [\tau] \rangle = \int_M \omega \wedge \tau, \tag{6.6}$$

which is well-defined: if $\omega = d\alpha$ with $\alpha \in \Omega^{k-1}(M)$ and $\tau \in Z_c^{n-k}(M)$, then

$$\int_M d\alpha \wedge \tau = \int_M d(\alpha \wedge \tau) - (-1)^{k-1} \int_M \alpha \wedge d\tau = \int_M d(\alpha \wedge \tau) = 0$$

by Stokes' theorem ($\alpha \wedge \tau$ has compact support since τ does), and the same argument applies in the second slot.

Theorem 6.8 (Poincaré duality)**.** *Let M be an oriented n-manifold admitting a finite good cover. Then* (6.6) *is nondegenerate: for every k the map*

$$H^k(M) \longrightarrow H_c^{n-k}(M)^\vee, \qquad [\omega] \longmapsto \big([\tau] \mapsto \textstyle\int_M \omega \wedge \tau\big), \tag{6.7}$$

is an isomorphism. In particular, if M is compact, then $H^k(M) \cong H^{n-k}(M)^\vee$ and $\dim H^k(M) = \dim H^{n-k}(M)$ for all k.

Proof. The proof is an induction on the size of a finite good cover, with the Mayer–Vietoris sequence and the five lemma as the induction step; it follows Bott–Tu [1, §5]. The sign bookkeeping deserves to be written out once and for all.

Signs and the adjunction identities. Write δ for the connecting map of the ordinary Mayer–Vietoris sequence (4.14) and δ_c for that of (6.2): for $\gamma \in Z^k(U \cap V)$ and $\tau \in Z_c^q(M)$,

$$\delta[\gamma] = [d\rho_V \wedge \gamma], \qquad \delta_c[\tau] = [d\rho_U \wedge \tau].$$

For the inclusions $i_U : U \to M$ etc. and their extension maps $(i_U)_*$, one has, with γ of degree k,

$$\big\langle i_U^*[\omega], [\tau] \big\rangle = \big\langle [\omega], (i_U)_*[\tau] \big\rangle, \tag{6.8}$$

$$\big\langle \delta[\gamma], [\tau] \big\rangle = (-1)^{k+1} \big\langle [\gamma], \delta_c[\tau] \big\rangle. \tag{6.9}$$

Indeed (6.8) is the identity $\int_U (i_U^*\omega) \wedge \tau = \int_M \omega \wedge (i_U)_*\tau$, and for (6.9):

$$\langle \delta[\gamma], [\tau] \rangle = \int_M d\rho_V \wedge \gamma \wedge \tau = (-1)^k \int_{U \cap V} \gamma \wedge d\rho_V \wedge \tau = (-1)^{k+1} \int_{U \cap V} \gamma \wedge d\rho_U \wedge \tau = (-1)^{k+1} \langle [\gamma], \delta_c[\tau] \rangle.$$

To absorb the sign $(-1)^{k+1}$ we twist the pairing: define

$$\varphi_M^k : H^k(M) \longrightarrow H_c^{n-k}(M)^\vee, \qquad \varphi_M^k([\omega])([\tau]) = (-1)^{\frac{k(k+1)}{2}} \int_M \omega \wedge \tau. \tag{6.10}$$

Since $\varepsilon_{k+1}/\varepsilon_k = (-1)^{k+1}$ for $\varepsilon_k = (-1)^{k(k+1)/2}$, the identities (6.8)–(6.9) imply that the ladders below commute, and because $\varepsilon_k \neq 0$, the twisted maps φ_M^k are isomorphisms if and only if the untwisted ones (6.7) are.

Base of the induction. $M = \mathbb{R}^n$. For $k = 0$, $H^0(\mathbb{R}^n) = \mathbb{R}[1]$ and $\varphi^0([1])([\tau]) = \int_{\mathbb{R}^n} \tau$, which is nonzero on the generator $[\mu]$ of $H_c^n(\mathbb{R}^n)$ (Remark 6.6); both spaces are 1-dimensional, so φ^0 is an isomorphism. For $k \geq 1$ both sides vanish.

Induction step. Let $M = U \cup V$ with $U = U_0 \cup \cdots \cup U_{m-1}$, $V = U_m \cong \mathbb{R}^n$, so that U and $U \cap V$ admit good covers with at most $m-1$ sets. The ordinary Mayer–Vietoris sequence (4.13) and the dual of the compact one (6.2) (duals of exact sequences of vector spaces are exact) form a ladder; writing φ_W^j for the twisted pairing map of the open set W in degree j:

$$\begin{tikzcd}
H^{k-1}(U\cap V) \arrow[r] \arrow[d, "\varphi^{k-1}_{U\cap V}"] & H^k(M) \arrow[r] \arrow[d, "\varphi^k_M"] & H^k(U)\oplus H^k(V) \arrow[r] \arrow[d, "\varphi^k_U\oplus\varphi^k_V"] & H^k(U\cap V) \arrow[r] \arrow[d, "\varphi^k_{U\cap V}"] & H^{k+1}(M) \arrow[d, "\varphi^{k+1}_M"] \\
H_c^{n-k+1}(U\cap V)^\vee \arrow[r] & H_c^{n-k}(M)^\vee \arrow[r] & H_c^{n-k}(U)^\vee\oplus H_c^{n-k}(V)^\vee \arrow[r] & H_c^{n-k}(U\cap V)^\vee \arrow[r] & H_c^{n-k-1}(M)^\vee
\end{tikzcd} \tag{6.11}$$

in which every square commutes: the squares involving restrictions by (6.8), the squares involving connecting maps by (6.9) together with the choice of ε_k. By induction on m, the maps $\varphi_{U\cap V}$, φ_U, φ_V are isomorphisms in all degrees. We prove φ_M^k is an isomorphism for all k by descending induction on k, starting from $k=n+1$ where both entries vanish. For given k, the vertical maps at positions $1,3,4$ are isomorphisms, and the map at position 5, φ_M^{k+1}, is an isomorphism by the descending induction. The five lemma yields φ_M^k an isomorphism.

Conclusion. For M compact, $\Omega_c^k=\Omega^k$, and (6.7) becomes $H^k(M)\cong H^{n-k}(M)^\vee$. The dimension statement follows because de Rham cohomology of a compact manifold is finite-dimensional (Corollary 5.10). $\square$

Remark 6.9. Poincaré duality holds for every oriented manifold, with or without a finite good cover; the general case follows from the finite one by a direct limit over finite good covers, see Bott–Tu [1, §5]. The orientation hypothesis is essential: it enters through the existence of an orientation class in $H_c^n(M)$ paired with $H^0(M)$; see also Proposition 7.12.

Example 6.10. (a) For S^n: the pairing sends $[1]\in H^0(S^n)$ and $[\mathrm{Vol}]\in H^n(S^n)$ to $\int_{S^n}\mathrm{Vol}\neq 0$; duality is the statement that these two classes are the only nonzero ones. (b) For $\mathbb{T}^2=S^1\times S^1$ with angular forms $d\theta_1, d\theta_2$: the classes $[1], [d\theta_1/2\pi], [d\theta_2/2\pi], [d\theta_1\wedge d\theta_2/(2\pi)^2]$ are a basis, and $\int_{\mathbb{T}^2} d\theta_1\wedge d\theta_2/(2\pi)^2=1$; duality pairs $[d\theta_1/2\pi]$ with $[d\theta_2/2\pi]$ (up to the sign $(-1)^{1\cdot 1}$) and $[1]$ with the volume class. (c) For $M=\mathbb{R}^n\setminus\{0\}$: Example 6.7 and Section 4.4 show H^0, H^{n-1} and H_c^1, H_c^n are the only nonzero groups, exactly as duality requires.

6.4 The Künneth formula

The wedge product gives a canonical map

$$\Phi:\bigoplus_{p+q=k} H^p(M)\otimes H^q(F)\longrightarrow H^k(M\times F),\qquad [\omega]\otimes[\eta]\longmapsto[\pi_1^*\omega\wedge\pi_2^*\eta]. \tag{6.12}$$

It is well-defined (closed forms map to closed forms; exactness is killed in each slot because π_i^* commutes with d and the wedge of an exact form with a closed one is exact).

Theorem 6.11 (Künneth formula for de Rham cohomology)**.** *If M admits a finite good cover, then* (6.12) *is an isomorphism for every manifold F and every k.*

Proof. Induct on the number m of sets in a finite good cover of M.

Base. $M=\mathbb{R}^n$ is contractible: $H^p(M)$ is $\mathbb{R}$ at $p=0$ and 0 otherwise, so the left-hand side of (6.12) is $H^k(F)$ via $[\omega]\otimes[\eta]\mapsto c\,[\eta]$ (where $\omega=c\cdot 1$). The right-hand side is $H^k(M\times F)$, and Φ sends $[1]\otimes[\eta]$ to $[\pi_2^*\eta]$; since $\pi_2: M\times F\to F$ is a homotopy equivalence, π_2^* is an isomorphism, so Φ is an isomorphism.

Induction step. Write $M=U\cup V$ with U covered by $U_0,\dots,U_{m-1}$ and $V=U_m$, so that U and $U\cap V$ have good covers with at most $m-1$ sets. Then $\{U\times F,\, V\times F\}$ is an open cover of $M\times F$ with intersection $(U\cap V)\times F$. To keep the notation readable, write $G^j(W)=H^j(W\times F)$

and $K^j(W) = \bigoplus_{p+q=j} H^p(W) \otimes H^q(F)$. The Mayer–Vietoris sequences then fit into a ladder (vertical maps: the Φ's)

$$\begin{tikzcd} G^{k-1}(U\cap V) \arrow[r] \arrow[d] & G^k(M) \arrow[r] \arrow[d] & G^k(U)\oplus G^k(V) \arrow[r] \arrow[d] & G^k(U\cap V) \arrow[r] \arrow[d] & G^{k+1}(M) \arrow[d] \\ K^{k-1}(U\cap V) \arrow[r] & K^k(M) \arrow[r] & K^k(U)\oplus K^k(V) \arrow[r] & K^k(U\cap V) \arrow[r] & K^{k+1}(M) \end{tikzcd} \tag{6.13}$$

The bottom row is the Mayer–Vietoris sequence of the cover $\{U, V\}$ tensored with $H^q(F)$ and summed over $p + q = k$; tensoring with a vector space is exact, so it is exact. The vertical maps are the Φ's of the various manifolds, and the squares commute: the restriction squares by naturality of pullbacks and wedges, and the connecting-map squares because $\delta_{M\times F}[\pi_1^*\gamma \wedge \pi_2^*\eta] = [\,d(\rho_V \circ \pi_1) \wedge \pi_1^*\gamma \wedge \pi_2^*\eta\,] = \Phi\big(\delta[\gamma] \otimes [\eta]\big)$ (the same convention (4.14) is used on both rows, so no sign appears). By induction, $\Phi_U, \Phi_V, \Phi_{U\cap V}$ are isomorphisms in all degrees; by descending induction on k (starting where both rows vanish, above $\dim M + \dim F$), the five lemma gives that Φ_M is an isomorphism in every degree. $\square$

Corollary 6.12. *$H^k(\mathbb{T}^n) \cong \Lambda^k\mathbb{R}^n$ for every k: if $\theta_1, \dots, \theta_n$ are the angular coordinates, the classes $[d\theta_{i_1}/2\pi \wedge \cdots \wedge d\theta_{i_k}/2\pi]$, $i_1 < \cdots < i_k$, form a basis of $H^k(\mathbb{T}^n)$.*

Proof. Iterate Theorem 6.11 with $M = S^1$, $F = \mathbb{T}^{n-1}$, using $H^*(S^1) = \mathbb{R}[1] \oplus \mathbb{R}[d\theta/2\pi]$ (Proposition 4.10); the wedge of pullbacks is the wedge of the angular forms. $\square$

7 Computations and applications

7.1 Spheres

Proposition 7.1. *For $n \geq 1$:*

$$H^k(S^n) = \begin{cases} \mathbb{R} & k \in \{0, n\}, \\ 0 & \textit{otherwise.} \end{cases}$$

The generator of $H^n(S^n)$ is the class of any volume form with nonzero integral, and the cup product $a \cup a = 0$ for $a \in H^n(S^n)$ by degree reasons ($2n > n$).

Proof. Induct on n, the case $n = 1$ being Proposition 4.10. Write $S^n = U \cup V$ with $U = S^n \setminus \{N\}$, $V = S^n \setminus \{S\}$ the stereographic charts; both are diffeomorphic to $\mathbb{R}^n$, and $U \cap V \simeq S^{n-1}$. For $k \geq 2$ the Mayer–Vietoris sequence (4.13) gives

$$H^{k-1}(U) \oplus H^{k-1}(V) \to H^{k-1}(U \cap V) \to H^k(S^n) \to H^k(U) \oplus H^k(V)$$

with the flanking terms zero, hence $H^k(S^n) \cong H^{k-1}(U \cap V) \cong H^{k-1}(S^{n-1})$. For $k = 1$ the sequence reads

$$H^0(S^n) \to H^0(U) \oplus H^0(V) \to H^0(U \cap V) \to H^1(S^n) \to H^1(U) \oplus H^1(V),$$

i.e. $\mathbb{R} \to \mathbb{R} \oplus \mathbb{R} \xrightarrow{(a,b)\mapsto a-b} \mathbb{R} \to H^1(S^n) \to 0$, and the difference map is surjective, so $H^1(S^n) = 0$. Together with $H^0(S^n) = \mathbb{R}$ and the induction hypothesis this proves the claim. $\square$

Remark 7.2. The vanishing of $a \cup a$ is the first visible instance of the general fact that de Rham cohomology is *graded commutative*: for $\alpha \in H^p, \beta \in H^q$, $\alpha \cup \beta = (-1)^{pq}\beta \cup \alpha$. Odd-dimensional classes therefore satisfy $a \cup a = -a \cup a = 0$; even-dimensional ones need not, as $H^*(\mathbb{C}\mathrm{P}^n)$ below shows.

7.2 Tori

Proposition 7.3. *The cohomology ring of the torus is the exterior algebra $H^*(\mathbb{T}^n) \cong \Lambda^*(\mathbb{R}^n)$: writing $a_i = [d\theta_i/2\pi]$,*

$$H^k(\mathbb{T}^n) = \bigoplus_{i_1<\cdots<i_k} \mathbb{R} \cdot (a_{i_1} \cup \cdots \cup a_{i_k}).$$

Proof. This is Corollary 6.12 together with the ring structure of the wedge product (which is the cup product by definition). □

Remark 7.4. The same ring distinguishes tori of different dimensions and, say, $\mathbb{T}^2$ from S^2: no homotopy equivalence between them exists. In fact the wedge-square map $H^1 \times H^1 \to H^2$ on $\mathbb{T}^2$ is a perfect pairing, the shadow of Poincaré duality.

7.3 Complex projective spaces and the Fubini–Study class

Theorem 7.5. *The de Rham cohomology ring of $\mathbb{C}\mathrm{P}^n$ is a truncated polynomial ring:*

$$H^*(\mathbb{C}\mathrm{P}^n) \cong \mathbb{R}[x]/(x^{n+1}), \qquad \deg x = 2.$$

The generator x is represented by the Fubini–Study form*: on the affine chart $\mathbb{C}^n = \{[1 : z_1 : \cdots : z_n]\}$,*

$$\omega_{\mathrm{FS}} = \frac{i}{2\pi}\,\partial\bar\partial \log\bigl(1+|z|^2\bigr) = \frac{i}{2\pi}\left[\frac{\sum_j dz_j \wedge d\bar z_j}{1+|z|^2} - \frac{\bigl(\sum_j \bar z_j\, dz_j\bigr) \wedge \bigl(\sum_k z_k\, d\bar z_k\bigr)}{(1+|z|^2)^2}\right]. \tag{7.1}$$

Proof. Step 1: the form is global and closed. Here $\partial = \sum_j \frac{\partial}{\partial z_j} dz_j \wedge$ and $\bar\partial = \sum_j \frac{\partial}{\partial \bar z_j} d\bar z_j \wedge$ are the Dolbeault operators of the chart. On the overlap of the charts $z_0 = 1$ and $z_1 = 1$ the coordinates are related by $w_1 = 1/z_1$, $w_j = z_j/z_1$ for $j \geq 2$, so that $1+|z|^2 = (1+|w|^2)/|w_1|^2$ and

$$\partial\bar\partial \log(1+|z|^2) = \partial\bar\partial \log(1+|w|^2) - \partial\bar\partial \log|w_1|^2 = \partial\bar\partial \log(1+|w|^2),$$

because $\log|w_1|^2 = \log w_1 + \log \bar w_1$ is annihilated by $\partial\bar\partial$ (holomorphic and antiholomorphic functions have no mixed term). Hence (7.1) defines one global form; it is real ($\overline{\omega_{\mathrm{FS}}} = \omega_{\mathrm{FS}}$) and closed, being $\partial\bar\partial$ of a function (indeed $\omega_{\mathrm{FS}} = d\alpha$ locally with $\alpha = (i/4\pi)(\bar\partial - \partial)\log(1+|z|^2)$).

Step 2: the integral over a projective line. Restrict to $\mathbb{C}\mathrm{P}^1 \subset \mathbb{C}\mathrm{P}^n$, the line $z_2 = \cdots = z_n = 0$. In the coordinate $z_1 = z = re^{i\theta}$:

$$\omega_{\mathrm{FS}}\big|_{\mathbb{C}\mathrm{P}^1} = \frac{i}{2\pi}\frac{dz \wedge d\bar z}{(1+|z|^2)^2} = \frac{1}{\pi}\frac{r\,dr \wedge d\theta}{(1+r^2)^2},$$

and

$$\int_{\mathbb{C}\mathrm{P}^1} \omega_{\mathrm{FS}} = \frac{1}{\pi}\int_0^{2\pi}\int_0^{\infty} \frac{r}{(1+r^2)^2}\,dr\,d\theta = \frac{1}{\pi}\cdot 2\pi \cdot \frac{1}{2} = 1.$$

Step 3: the cohomology groups. Let $U = \mathbb{C}\mathrm{P}^n \setminus \{[1:0:\cdots:0]\}$ and $V = \mathbb{C}\mathrm{P}^n \setminus \mathbb{C}\mathrm{P}^{n-1} \cong \mathbb{C}^n$, where $\mathbb{C}\mathrm{P}^{n-1} = \{[0 : z_1 : \cdots : z_n]\}$. Then $U \cap V = \mathbb{C}^n \setminus \{0\} \simeq S^{2n-1}$, and U deformation retracts onto $\mathbb{C}\mathrm{P}^{n-1}$ via $[z_0 : z_1 : \cdots : z_n] \mapsto [tz_0 : z_1 : \cdots : z_n]$, $1 \geq t \geq 0$. The Mayer–Vietoris sequence, together with Proposition 7.1 for S^{2n-1} and the fact that V is contractible, gives, by induction on n (with $\mathbb{C}\mathrm{P}^1 = S^2$),

$$H^k(\mathbb{C}\mathrm{P}^n) = \begin{cases} \mathbb{R} & k \text{ even},\ 0 \leq k \leq 2n, \\ 0 & \text{otherwise.} \end{cases} \tag{7.2}$$

Indeed: for $k \notin \{0, 2n-1, 2n\}$ the sequence gives $H^k(\mathbb{C}\mathrm{P}^n) \cong H^k(U) \cong H^k(\mathbb{C}\mathrm{P}^{n-1})$; the connecting map $\delta : H^{2n-1}(U \cap V) \to H^{2n}(\mathbb{C}\mathrm{P}^n)$ is an isomorphism (its neighbours $H^{2n-1}(U), H^{2n-1}(V)$ vanish); and $H^{2n-1}(\mathbb{C}\mathrm{P}^n) \cong H^{2n-1}(U) = 0$.

Step 4: the ring structure. Put $x = [\omega_{\mathrm{FS}}] \in H^2(\mathbb{C}\mathrm{P}^n)$. Since $\int_{\mathbb{C}\mathrm{P}^1} x = 1$, the class x is nonzero, hence generates $H^2(\mathbb{C}\mathrm{P}^n) = \mathbb{R}$. For $k \le n$, the restriction of $\omega_{\mathrm{FS}}^{\wedge k}$ to $\mathbb{C}\mathrm{P}^k \subset \mathbb{C}\mathrm{P}^n$ is the k-fold wedge of the Fubini–Study form of $\mathbb{C}\mathrm{P}^k$, and the same computation as in Step 2, iterated, gives

$$\int_{\mathbb{C}\mathrm{P}^k} \omega_{\mathrm{FS}}^{\wedge k} = 1.$$

(The k-fold integral reduces to the one-variable integral $\int_0^\infty r^{2k-1}(1+r^2)^{-(k+1)}dr = 1/(2k)$ times the volume $2\pi^k/(k-1)!$ of S^{2k-1}, and $\omega_{\mathrm{FS}}^{\wedge k}$ carries the factor $k!$ from the wedge: $k! \cdot \pi^{-k} \cdot \frac{2\pi^k}{(k-1)!} \cdot \frac{1}{2k} = 1$.) Hence $x^k \neq 0$ for $k \le n$, so x^k generates $H^{2k}(\mathbb{C}\mathrm{P}^n) = \mathbb{R}$; and $x^{n+1} = 0$ since $H^{2n+2}(\mathbb{C}\mathrm{P}^n) = 0$. ☐

Remark 7.6. The class x is the Poincaré dual of the hyperplane $\mathbb{C}\mathrm{P}^{n-1}$: by the computation above, $\int_{\mathbb{C}\mathrm{P}^k} x^k = 1$ for every k, which is exactly the statement $\mathrm{PD}([\mathbb{C}\mathrm{P}^{n-1}]) = x$. This identification, plus the fact that ω_{FS} is the curvature form of a natural connection on the tautological line bundle, is the entry point to Chern–Weil theory.

7.4 Degree theory

Definition 7.7. Let $f : M \to N$ be a smooth map between compact, connected, oriented n-manifolds. The *degree* of f is

$$\deg f = \frac{\int_M f^*\omega}{\int_N \omega}$$

for any volume form ω on N with $\int_N \omega \neq 0$. This is independent of ω: any two volume forms differ by an exact form plus a scalar multiple, since $\dim H^n(N) = 1$ (Poincaré duality), and f^* annihilates exact forms.

Proposition 7.8. *(i)* $\deg(g \circ f) = \deg g \cdot \deg f$*; (ii)* $\deg \mathrm{id} = 1$*; (iii) homotopic maps have the same degree; (iv) if f is not surjective, then* $\deg f = 0$*; (v) an orientation-preserving diffeomorphism has degree* 1*; (vi) the antipodal map $A : S^n \to S^n$ has* $\deg A = (-1)^{n+1}$.

Proof. (i) and (ii) are immediate from $(g \circ f)^* = f^* \circ g^*$; (iii) is Theorem 4.1. For (iv), choose a volume form ω supported in $N \setminus f(M)$; then $f^*\omega = 0$. For (v), use Proposition 2.9. For (vi), write the standard volume form of S^n as $\omega = \iota_\nu(dx^1 \wedge \cdots \wedge dx^{n+1})$ where $\nu = \sum_i x^i\partial_i$ is the outward unit normal (the restriction of the ambient form to S^n). Since $dA_p = -\mathrm{id}$ on $\mathbb{R}^{n+1}$ and the outward normal at $A(p)$ is $\nu_{A(p)} = -\nu_p$, one has $A^*\omega = (-1)^{n+1}\omega$; integrating, $\deg A = (-1)^{n+1}$. ☐

7.5 Three classical applications

Theorem 7.9 (Brouwer fixed-point theorem)**.** *Every continuous map $f : \overline{D^n} \to \overline{D^n}$ of the closed n-ball has a fixed point.*

Proof. It suffices to treat smooth f: a continuous f with no fixed point has $\min_x |f(x) - x| = \delta > 0$, and a smooth approximation within $\delta/2$ again has no fixed point; a fixed point of the approximation would force one of f by passing to a limit.

Suppose f has no fixed point. For each $x \in \overline{D^n}$ let $r(x)$ be the point where the ray from $f(x)$ through x meets $S^{n-1} = \partial D^n$; since $x \neq f(x)$ and $x, f(x) \in D^n$, the ray meets the sphere in a unique point, and $r : \overline{D^n} \to S^{n-1}$ is smooth. If $x \in S^{n-1}$ then $r(x) = x$: r is a retraction. With $i : S^{n-1} \hookrightarrow D^n$ the inclusion, $r \circ i = \mathrm{id}_{S^{n-1}}$, hence $\mathrm{id} = i^* \circ r^*$ on $H^{n-1}(S^{n-1})$. But $H^{n-1}(S^{n-1}) = \mathbb{R}$ (Proposition 7.1) while $H^{n-1}(D^n) = 0$ (D^n is contractible), so $r^* = 0$ and $i^* \circ r^* = 0 \neq \mathrm{id}$: contradiction. ☐

Theorem 7.10 (Hairy ball theorem)**.** *There is no nonvanishing continuous vector field on S^{2m}.*

Proof. A continuous nonvanishing field can be smoothed and normalized, so suppose $v : S^{2m} \to \mathbb{R}^{2m+1}$ is smooth with $|v(p)| = 1$ and $v(p) \perp p$ for all p. Then $F(p,t) = \cos(\pi t)\, p + \sin(\pi t)\, v(p)$ is a smooth homotopy from $\mathrm{id}_{S^{2m}}$ to the antipodal map A (the sum of two perpendicular unit vectors of squared lengths $\cos^2 + \sin^2 = 1$ lies on the sphere). By Proposition 7.8(iii) and (vi), $1 = \deg \mathrm{id} = \deg A = (-1)^{2m+1} = -1$, a contradiction. □

Theorem 7.11 (Fundamental theorem of algebra)**.** *Every nonconstant complex polynomial has a root in* $\mathbb{C}$*.*

Proof. Let $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$, $n \geq 1$, and suppose p has no root. For $R \geq 0$ define $f_R : S^1 \to S^1$, $f_R(z) = p(Rz)/|p(Rz)|$; this is well-defined and depends smoothly on R, so all f_R are mutually homotopic and have the same degree. Now f_0 is the constant map $z \mapsto p(0)/|p(0)|$, of degree 0 (Proposition 7.8(iv), or directly). On the other hand,

$$f_R(z) = \frac{z^n + R^{-1}a_{n-1}z^{n-1} + \cdots + R^{-n}a_0}{\left|z^n + R^{-1}a_{n-1}z^{n-1} + \cdots + R^{-n}a_0\right|}$$

converges uniformly to $z \mapsto z^n$ as $R \to \infty$; uniformly close maps are homotopic (through the normalized linear homotopy), so for R large, $\deg f_R = \deg(z \mapsto z^n) = n$ (the map $z \mapsto z^n$ pulls the angular form back to n times itself). Hence $0 = n$, contradicting $n \geq 1$. □

7.6 Orientability and dimension

Proposition 7.12. *If M is a closed, connected, nonorientable n-manifold, then $H^n(M) = 0$. (For comparison, an orientable closed M has $H^n(M) = \mathbb{R}$ by Poincaré duality.)*

Proof. Let $\pi : \widehat{M} \to M$ be the orientation double cover: its points are pairs (p, o) of a point and a local orientation at p, topologized and smooth in the evident way, so that π is a 2-sheeted covering and the nontrivial deck transformation $\sigma : \widehat{M} \to \widehat{M}$ reverses orientation. The manifold $\widehat{M}$ is compact, connected and orientable, so $H^n(\widehat{M}) = \mathbb{R}$, generated by a volume form Vol; since σ reverses orientation, $\int_{\widehat{M}} \sigma^* \mathrm{Vol} = -\int_{\widehat{M}} \mathrm{Vol}$ (Proposition 2.9), hence $\sigma^*[\mathrm{Vol}] = -[\mathrm{Vol}]$ in $H^n(\widehat{M}) = \mathbb{R}$.

Let $\omega \in Z^n(M)$. Then $\pi^*\omega \in Z^n(\widehat{M})$, so $[\pi^*\omega] = c\,[\mathrm{Vol}]$ for some $c \in \mathbb{R}$. Since $\sigma^* \circ \pi^* = \pi^*$ (as $\pi \circ \sigma = \pi$), $c\,[\mathrm{Vol}] = \sigma^*[\pi^*\omega] = c\,\sigma^*[\mathrm{Vol}] = -c\,[\mathrm{Vol}]$, hence $c = 0$ and $[\pi^*\omega] = 0$: say $\pi^*\omega = d\eta$. Average under the deck group: $\eta' = \frac{1}{2}(\eta + \sigma^*\eta)$ satisfies $\sigma^*\eta' = \eta'$, so $\eta' = \pi^*\tau$ for a unique $\tau \in \Omega^{n-1}(M)$. Then $\pi^*(d\tau) = d(\pi^*\tau) = d\eta' = \frac{1}{2}(d\eta + \sigma^* d\eta) = \pi^*\omega$, and since π^* is injective, $d\tau = \omega$. Thus $[\omega] = 0$ for every closed ω, i.e. $H^n(M) = 0$. □

Corollary 7.13 (Invariance of dimension)**.** *If $\mathbb{R}^n$ and $\mathbb{R}^m$ are homeomorphic, then $n = m$.*

Proof. A homeomorphism $\mathbb{R}^n \to \mathbb{R}^m$ restricts to a homeomorphism $\mathbb{R}^n \setminus \{0\} \to \mathbb{R}^m \setminus \{0\}$, which induces an isomorphism of singular, hence (Theorem 5.8) of de Rham cohomology. Now $\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}$, so $H^{n-1}(\mathbb{R}^n \setminus \{0\}) = \mathbb{R}$ by Proposition 7.1. If $n \geq 2$ and $m \geq 2$, the isomorphism forces $H^{n-1}(\mathbb{R}^m \setminus \{0\}) = \mathbb{R}$, hence $n - 1 \in \{0, m-1\}$, i.e. $n = m$ (the case $n = m = 1$ is the identity). The mixed cases are excluded by H^0: $\mathbb{R} \setminus \{0\}$ has two components while $\mathbb{R}^m \setminus \{0\}$ has one for $m \geq 2$. □

Exercise 7.14. (a) Show $H^*(\mathbb{R}\mathrm{P}^2) = \mathbb{R}$ in degree 0 and 0 otherwise, using Proposition 7.12, the Mayer–Vietoris decomposition $\mathbb{R}\mathrm{P}^2 = U \cup V$ with U a Möbius band (homotopy equivalent to S^1) and V a disk, and the fact that $H^2(\mathbb{R}\mathrm{P}^2) = 0$. What is the image of $H^1(\mathbb{R}\mathrm{P}^2) \to H^1(U) = \mathbb{R}$?
(b) Let $\pi : S^n \to \mathbb{R}\mathrm{P}^n$ be the antipodal quotient and A the antipodal map. Show first that $\pi^* : H^k(\mathbb{R}\mathrm{P}^n) \to H^k(S^n)$ is injective: if $\pi^*\omega = d\eta$, average $\eta' = \frac{1}{2}(\eta + A^*\eta)$ and descend (η' is A-invariant, hence $\eta' = \pi^*\tau$ for a unique τ). Conclude that $H^k(\mathbb{R}\mathrm{P}^n) = 0$ for $1 \leq k \leq n-1$ by Proposition 7.1.

Remark 7.15. The applications collected here barely scratch the surface. The same circle of ideas gives: the Borsuk–Ulam theorem; the Lefschetz fixed-point formula (which, for a compact manifold, computes the intersection number of the graph with the diagonal in $M \times M$ via the cohomology of M); the Hopf degree theorem (maps $S^n \to S^n$ are classified by degree); and the Gauss–Bonnet–Chern theorem already mentioned in Remark 5.11. Bott–Tu [1, §11, §17] and Madsen–Tornehave [5, Ch. 16–18] treat most of these with exactly the tools developed here.

8 Further directions

The de Rham complex is the seed of a large part of modern geometry. This closing chapter collects, without proofs, the four directions most directly visible from what has been done here.

8.1 The sheaf-theoretic viewpoint

Much of Chapters 5–6 can be reorganized into one statement about sheaves. Let $\underline{\mathbb{R}}$ be the constant sheaf on M and $\underline{\Omega}^k$ the sheaf of smooth k-forms. The Poincaré lemma says precisely that

$$0 \longrightarrow \underline{\mathbb{R}} \longrightarrow \underline{\Omega}^0 \xrightarrow{d} \underline{\Omega}^1 \xrightarrow{d} \underline{\Omega}^2 \longrightarrow \cdots \tag{8.1}$$

is a *resolution* of $\underline{\mathbb{R}}$ (exact as a sequence of sheaves). The sheaves $\underline{\Omega}^k$ are *soft*: sections over closed sets extend, as one sees with bump functions—the argument of Remark 5.4. Soft sheaves have vanishing higher cohomology, so (8.1) is an acyclic resolution, and the abstract de Rham theorem (Warner [3, §5]) gives

$$H^k(M; \underline{\mathbb{R}}) \cong H^k\big(\Omega^*(M), d\big) = H^k_{\mathrm{dR}}(M), \tag{8.2}$$

recovering Theorem 5.8 for sheaf cohomology. This reformulation is what transports the subject to complex and algebraic geometry: replace $\underline{\Omega}^k$ by the sheaves of holomorphic forms, or of algebraic differentials, and (8.2) becomes the definition of Dolbeault and algebraic de Rham cohomology. The double complex of Section 5.2, in turn, is the standard Čech-resolution comparison, and its two filtrations are spectral sequences in embryonic form.

8.2 Hodge theory

Fix a Riemannian metric on a compact oriented manifold M. The metric induces a pointwise inner product on forms and a formal adjoint $d^* : \Omega^{k+1} \to \Omega^k$; the Hodge Laplacian $\Delta = dd^* + d^*d$ is elliptic. Hodge's theorem states that

$$\Omega^k(M) = \mathcal{H}^k(M) \oplus \operatorname{im} d \oplus \operatorname{im} d^*,$$

where $\mathcal{H}^k(M) = \ker \Delta$ is the finite-dimensional space of *harmonic* forms. Since $\ker \Delta = \ker d \cap \ker d^*$ (an integration by parts), the composite $\mathcal{H}^k(M) \hookrightarrow Z^k(M) \to H^k(M)$ is an isomorphism: every cohomology class has a unique harmonic representative. This gives a new proof of finite-dimensionality, an inner product on cohomology, and the star operator $\star$ (defined by $\alpha \wedge \star\beta = \langle \alpha, \beta \rangle \,\mathrm{Vol}$) exchanging $\mathcal{H}^k$ and $\mathcal{H}^{n-k}$—a metric refinement of Poincaré duality. Bott–Tu prove Hodge theory from scratch in their Chapter 1; Warner [3, Ch. 6] gives the elliptic proof.

8.3 Chern–Weil theory

If $E \to M$ is a vector bundle with connection ∇ and curvature F_∇, then for every invariant polynomial φ the form $\varphi(F_\nabla)$ is closed, and its class in $H^*(M)$ is independent of the connection. This is the Chern–Weil theorem, and it produces the Chern classes $c_k(E) \in H^{2k}(M)$ from the differential geometry of connections. The Fubini–Study form of Section 7.3 is exactly this construction for the tautological line bundle over $\mathbb{CP}^n$: ω_{FS} is (up to constants) the curvature of the natural metric connection, and x is its first Chern class. Combined with de Rham's theorem, Chern–Weil closes the circle begun in Remark 5.11: the Gauss–Bonnet–Chern theorem is the special case $\varphi = \mathrm{Pf}$.

8.4 Cohomology operations and beyond

Two structural remarks to finish. First, the de Rham complex carries more than its cohomology: the graded-commutative algebra structure on forms descends to H^*, but so does, for instance, the *Massey product* structure under suitable vanishing hypotheses—the first place where the form-level algebra strictly refines the cohomological one (see Bott–Tu [1, §19] on "the Massey triple product": the same homotopy theory as the cylinder operators of Chapter 4, iterated). Second, all arguments of these notes are real; integral information (torsion) is invisible to forms, and recapturing it—by flat connections and local systems, by the Hodge-theoretic integral lattices, or by differential cohomology—remains an active theme. The reader who has absorbed the present chapters is equipped for all three.

Remark 8.1 (Sources). The organization of Chapters 4–6 follows Bott–Tu [1], with the following deviations: the invariant formula for d and its verification (Chapter 3) follow the uniqueness argument of Warner [3] but are proved in coordinates; Lemma 5.5 makes explicit the spectral-sequence degenerations that Bott–Tu quote; the sign corrections in the Poincaré duality induction (Section 6.3) are written out rather than absorbed into a convention. The manifold-theoretic prerequisites of Chapter 2 follow Lee [2] and Tu [4]; the computations of Section 7 are standard, with the Fubini–Study integral written out in full.

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